

God*: A Role-Based Definition of the Ground of the Physical Order

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The Ground of the Physical Order as a Shared Vocabulary: Role, Occupant, and Four Verdicts for Comparing Metaphysical Theories across Creeds

A monograph in the form of a doctoral dissertation

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“The first question which we have a right to ask will be: Why is there something rather than nothing?” — G. W. Leibniz, *Principles of Nature and Grace* §7 (1714)

“Not *how* the world is, is the mystical, but *that* it is.” — Ludwig Wittgenstein, *Tractatus Logico-Philosophicus* 6.44 (1921)

“The eternal mystery of the world is its comprehensibility.” — Albert Einstein, “Physics and Reality” (1936)

The first two epigraphs ask about G-existence: that there is a world at all. The third asks about G-law: that the world is orderly enough to understand. This thesis is about the single question that sits above both.

Abstract

This thesis defends a role-based definition of the word “God,” written **God***, so that believers, atheists and people outside the Abrahamic traditions can argue about one precisely stated question instead of past each other. God* is not the name of a being. It is the title of a role: *the non-internal ground of the physical order as a whole*, where that whole has two sides — **G-existence** (that there is a physical domain at all) and **G-law** (that

the domain is lawful). A theory of reality then gives one of four verdicts: **Filled** (one non-internal ground of both sides), **Split** (the sides have different grounds, or only one side has one), **Empty** (neither side has a non-internal ground), or **Refused** (the role question rests on a false presupposition).

Part I is a first-principles course in the metaphysics a reader needs: names and titles, roles and occupants, explanation, causation and grounding, necessity and brute fact, the principle of sufficient reason, theories of the laws of nature, and what cosmology can and cannot settle. It also traces the question of an ultimate ground from the Presocratics to present-day analytic philosophy, across Greek, Abrahamic, Indian and Chinese thought.

Part II states the framework formally, in the style of Ramsey and Lewis, proves that its verdicts are exhaustive and exclusive, and places it beside its nearest ancestors: Nelson Pike on “God” as a title, the Comparative Religious Ideas Project, John Hick’s “the Real,” J. L. Schellenberg’s “ultimism,” and Graham Oppy’s worldview comparison. The claim of originality is modest and specific: the two-sided role, the Split verdict, the presupposition test that separates Refused from Empty, and the rule that verdicts score *claims* rather than traditions.

Part III resolves the framework’s hardest problems:

1. **What “ground” means.** It becomes a generic ultimate-explanation relation with four species: productive, constitutive, nomic and axiological.
2. **Hard cases.** Simulators, Spinoza, priority monism, a necessary cosmos, Advaita Vedānta, axiarchism, idealism, the mathematical universe, process theism and others each get a determinate verdict.
3. **Refused versus Empty.** They are separated by three stated presuppositions.
4. **The scope of “physical.”** It is fixed by causal-nomic connection, not by the content of current physics, which avoids Hempel’s dilemma.

Part IV gives a scoring protocol, an atlas of more than forty views, the framework’s limits, and a revised statement, **God* v2**.

Reader’s guide

The thesis assumes no background. Three devices help.

- **Ladder boxes.** Where an idea is known to be hard, a boxed “Ladder” explains it four times, each rung harder than the last. **Rung 1** is for a curious ten-year-old. **Rung 2** is for a high-school student. **Rung 3** is for an undergraduate philosophy major. **Rung 4** is for a

graduate student or researcher. Climb until the rung feels too easy, then read the next one.

- **Primer pointers.** A name followed by $\rightarrow A.n$ (for example, *Plato* $\rightarrow A.3$) has a short primer in Appendix A. The main text uses each idea without stopping to teach its history.
- **Key-idea lines.** A line beginning **Key idea** states the one thing to carry forward from a section.

Notation.

- “God*” always means the role, never a particular god.
- **P** is the physical order.
- **E(P)** is the fact that P exists: G-existence.
- **L(P)** is the fact that P is lawful: G-law.
- “Ground*”, with a star, is the generic relation defined in Chapter 9.
- Verdicts are written F, S, E, R. Subscripts mark what kind of ground is involved, for example F_p for a productive ground.

How to read it.

- A reader who wants only the framework can read Chapters 6, 7 and 16 and Appendix B.
- A reader who wants the lesson should go in order.
- A reader who wants to know whether the idea can be published should read Chapter 8 and Appendix D.

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Part I — Foundations: a first-principles lesson in metaphysics

Part I teaches, from scratch, every idea the God* framework uses. Nothing in it is original. It is the ground floor the rest of the thesis stands on.

Chapter 1 — What metaphysics is, and why this question is hard

1.1 The word and the subject

“Metaphysics” began as a librarian’s label. Aristotle wrote a set of treatises on the most general features of reality (→ A.4). About three centuries later his editor, traditionally Andronicus of Rhodes, placed them *after* the *Physics*: in Greek, *ta meta ta phusika*. The label stuck, and it now names a subject.

Today metaphysics is the study of what there is and what it is like at the most general level. It asks what it is to exist, what makes a thing the same thing over time, what causation is, what makes something necessary or merely possible, and whether time passes. The question in this thesis belongs to its oldest and most ambitious branch, sometimes called *ultimate explanation*: why there is a world at all, and why it is the orderly kind of world it is.

Key idea. Metaphysics is not the study of the supernatural. It is the study of the most general structure of reality, whatever that structure turns out to be.

1.2 The ladder of “why”

Children discover the central problem of metaphysics by themselves. Ask “why?” after every answer and you climb a ladder. The glass broke because it fell. It fell because of gravity. Gravity works as it does because of the laws of general relativity. Then you ask why *those* laws hold, and finally why there is a lawful physical world at all. Figure 1.1 draws the climb.

Every rung but the last is answered from inside the physical order

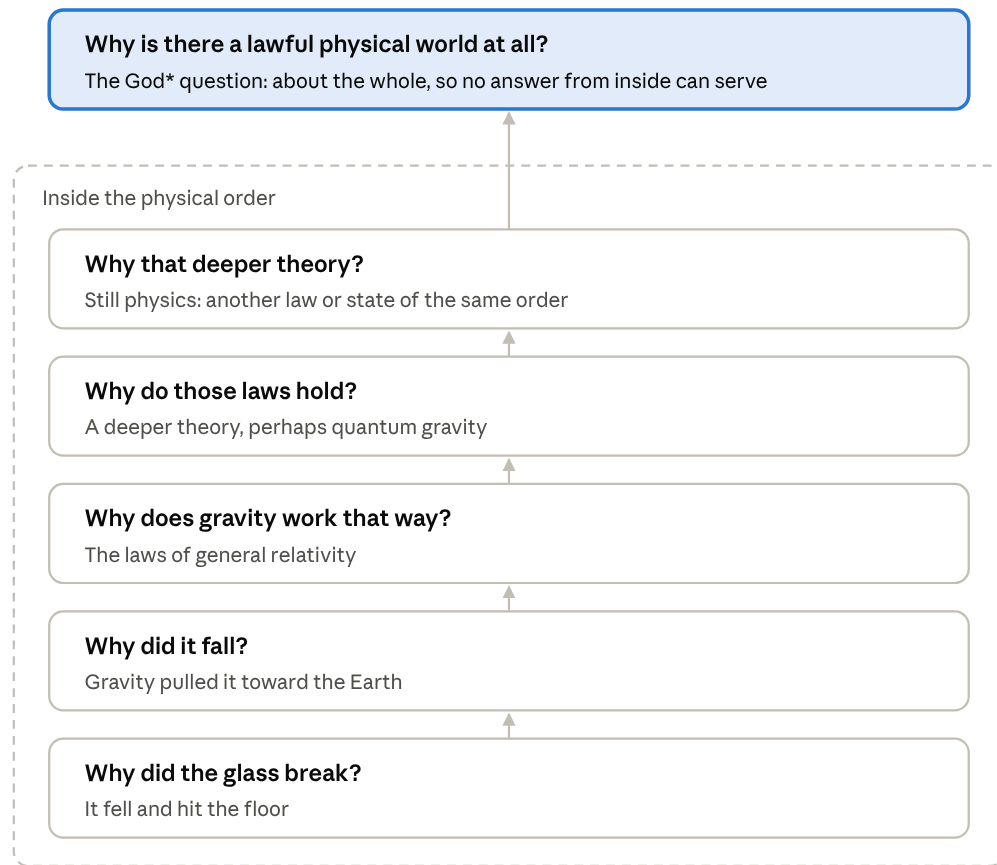


Figure 1.1 · The ladder of “why” questions

Each rung explains the one below by pointing to something more general. The lower rungs are answered from *inside* the physical world: one physical thing or law explains another. At the top, the question is about the physical world as a whole. Any answer drawn from inside the world would assume the very thing in question. The infographic calls those answers *internal*. The God* framework is a precise way of asking what, if anything, sits at the top.

1.3 Why the top question is hard

There are four separate difficulties.

1. **No outside vantage point.** Every experiment happens inside the physical world, so no experiment can observe the world from outside. This does not make the question meaningless, but it makes it a question for argument rather than measurement.

2. **Every chain of explanation must end somewhere.** The ancient skeptic Agrippa noticed that any chain of reasons has only three possible shapes (→ A.2). It goes on forever, it loops back on itself, or it stops. A stopping point is either a *brute fact* (true for no reason) or something that explains itself. These three endings reappear, in new clothes, as the verdicts of Chapter 7.
3. **Language strains.** Words like “nothing,” “before time” and “outside the universe” behave oddly. “Nothing” is not a very empty something. Much of the history of the question is a history of people tripping over such words.
4. **People mean different things by “God.”** A Christian theologian, a Hindu Vedāntin, a deist and a Stoic can all say “God exists” and mean four different claims. An atheist denying “God” may be denying a fifth. Philosophers call this *equivocation*: one word, several meanings, and an argument that slides between them. The God* framework exists to stop the sliding.

Ladder — Agrippa’s trilemma

Rung 1. If you keep asking “why?”, there are only three things that can happen. You can keep going forever, you can come back to an answer you already gave, or you can stop at an answer that has no answer behind it.

Rung 2. A reason needs support. Its support needs support. The chain is infinite (a regress), circular (A because B because A), or finite, ending in something unsupported. Each option looks unsatisfying at first, and philosophers still disagree about which is least bad.

Rung 3. In epistemology the trilemma drives the choice between infinitism, coherentism and foundationalism. In metaphysics it drives the choice between an infinite regress of explainers, self-explanation, and a brute or necessary terminus. Leibniz argued that even an infinite regress leaves the *whole series* unexplained (→ A.10).

Rung 4. The trilemma is exhaustive only for *linear* explanatory chains. Grounding theorists debate whether ground must be well-founded, a requirement defended by Schaffer and doubted by Bliss. Some also ask whether explanation can be holistic rather than chain-like. God* inherits this. A Filled verdict posits a terminus. Empty allows a brute terminus, a necessary terminus or a regress inside P. Refused may reject the chain model itself.

1.4 How metaphysics is done

Metaphysicians cannot run experiments on the whole, so they use four main methods.

- **Conceptual analysis.** Clarify what a concept requires. The God* definition is a piece of conceptual engineering in this tradition.
- **Inference to the best explanation.** Prefer the theory that explains the most with the least.
- **Theoretical virtues.** Weigh simplicity, scope, coherence and fit with science.
- **Thought experiments.** Imagine cases that test a definition, like the simulated universe in Chapter 10.

Metaphysics has had famous critics.

- **Hume** said claims that are neither mathematics nor experience should be “committed to the flames” (→ A.12).
- **Kant** argued that reason produces contradictions when it tries to reach beyond possible experience (→ A.13).
- **The logical positivists of the 1920s and 1930s** called such claims literally meaningless (→ A.21).

Metaphysics revived after the 1950s through Quine, Kripke and Lewis (→ A.22). The critics still matter here, because their positions are the clearest examples of the **Refused** verdict.

Chapter 2 — The toolkit: names, roles, explanation and dependence

2.1 Names, descriptions and titles

The definition’s first sentence is “God* is the title of a role, not the name of a being.” To see why that matters, compare three kinds of words that pick out things.

- **Names.** “Venus” picks out a planet directly. Gottlob Frege noticed that “the Morning Star” and “the Evening Star” pick out the same planet, Venus, while carrying different information (→ A.20). Saul Kripke later argued that proper names are *rigid*: “Venus” names the same object in every possible situation.
- **Definite descriptions.** “The tallest mountain” picks out whatever satisfies the description. Bertrand Russell showed that “the present King of France is bald” is not about a strange non-existent king. It says that exactly one thing is King of France and that thing is bald, so the sentence is false (→ A.20).

- **Titles.** “The President” names an office. The office can be filled by different people at different times, or be vacant. Nelson Pike argued in 1969 that “God” in classical theism works like a title: it names the office of the being that is omnipotent, omniscient and perfectly good, and “Yahweh” or “Allah” names the claimed occupant.

If “God” is a name, “God does not exist” is puzzling: what is the name naming? If it is a title, the sentence is simple. It says the office is vacant, just as “there is no King of France” does. That is why the infographic’s first footnote can say “Empty is a legal answer.”

Key idea. A title can be discussed by people who disagree about whether anyone holds it. That is what makes God* a *shared* vocabulary.

Ladder — name versus title

Rung 1. “Serena” is a name: it points at one person. “Class captain” is a job: anyone could have it, and some years nobody does.

Rung 2. You can argue about who the class captain is, or whether there is one, without first agreeing who it is. A job description lets people disagree about the answer while agreeing on the question.

Rung 3. Names are rigid designators. Titles are non-rigid descriptions or offices. Pike used this to address the old puzzle of whether God can sin. If “God” is a title that includes perfect goodness, then “God sins” is necessarily false, while the individual who holds the office might still have the *power* to sin.

Rung 4. Title-term readings raise hard questions about how the tetragrammaton and “Allāh” function in their home traditions: as names, titles, or both. God* avoids the dispute. It stipulates a new, starred technical term and leaves every tradition’s own word as it is. The occupant, if any, may later be named rigidly.

2.2 Roles and occupants

The idea of a role with possible occupants became a working tool in philosophy of mind.

Functionalism says that being in pain is occupying a certain causal role: caused by damage, causing distress and avoidance. What occupies the role in humans (certain nerve firings) could differ in an octopus or a robot. The heart is similar: “heart” names the pumping role, and an artificial pump can occupy it.

Frank Ramsey (→ A.20) and later David Lewis turned this into a general method for defining terms. Here is the method in three steps.

1. **Write the theory.** Collect what you believe about the thing, as a set of sentences that use the term.
2. **Replace the term with a blank.** The sentences now describe a job without saying who does it.
3. **The term names whatever uniquely fills the blank.** If exactly one thing does the job, the term refers to it. If nothing does, the term is empty. If several things do equally well, the term fails to pick out a unique referent.

The God* definition is a Ramsey–Lewis definition. Its “theory” is deliberately thin: only the ground-of-the-physical-order clauses, with no goodness, personhood or worship-worthiness. Lewis’s three outcomes, one occupant, none, or too many, become the Filled, Empty and Split verdicts. Figure 2.1 shows the idea.

One job description, three possible outcomes

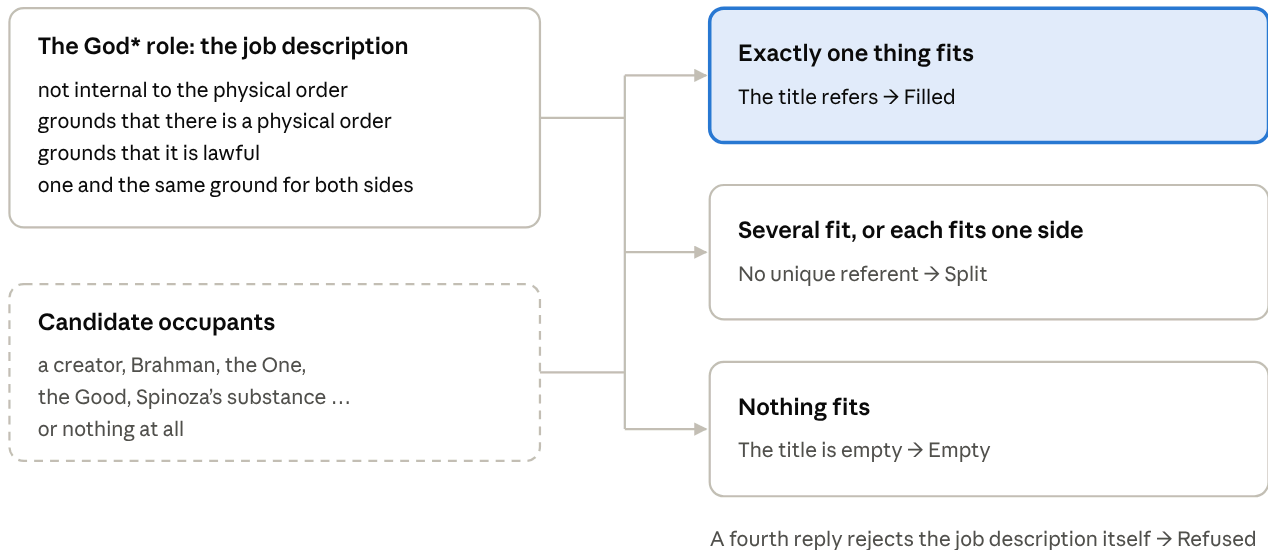


Figure 2.1 · Role and occupant, after Ramsey and Lewis

Ladder — the Ramsey-Lewis method

Rung 1. Write down everything a goalkeeper does, but leave the name blank.

Whoever does all of that is the goalkeeper.

Rung 2. Scientists did this with genes before anyone knew about DNA. “Gene” meant “whatever carries inherited traits in such-and-such a way.” Later they found that DNA sequences fill that role.

Rung 3. Take a theory $T(t)$ containing a new term t . Its *Ramsey sentence* is $\exists x T(x)$: something plays the T -role. Lewis adds uniqueness: t names the unique x such that $T(x)$, and fails to refer if there is none or more than one. Near-realizers, things that satisfy most of T , may be accepted when nothing satisfies all of it.

Rung 4. The Canberra Plan (Jackson 1998) generalizes the method. Collect the folk platitudes, Ramsify, then let science or metaphysics find the occupant. Critics question whether platitudes are stable, and whether the method gives the concept's users too much authority. God* sidesteps the first worry by stipulating a minimal role instead of collecting folk platitudes about God. It must still show that the role is *interesting*: that traditions actually care who fills it (Chapter 8).

2.3 What an explanation is

An explanation has two parts. The *explanandum* is what needs explaining. The *explanans* is what does the explaining. Explanations come in kinds. Aristotle distinguished four, often called his “four causes” ($\rightarrow$ A.4):

- the **material** cause: what a thing is made of;
- the **formal** cause: its structure or pattern;
- the **efficient** cause: what brought it about;
- the **final** cause: what it is for.

Modern philosophy adds that explanation is often **contrastive**. We explain why this happened *rather than* that. “Why is there something rather than nothing?” contrasts a world with no world, and matches G-existence. “Why is the world orderly rather than chaotic?” matches G-law. “Why these laws rather than others?” is a third contrast, which Derek Parfit called “Why this?” ($\rightarrow$ A.23). Chapter 13 decides how God* handles it.

2.4 Causation and grounding: two kinds of “because”

Compare two sentences:

- The window broke *because* the ball hit it.
- The ball is red *because* it is scarlet.

The first “because” is **causal**. One event produces a later one, across time. The second is not causal. Nothing happens first. Being scarlet simply *is one way* of being red, and the ball’s redness holds *in virtue of* its scarlet colour. Since about 2009 metaphysicians have called this second relation **grounding**. Standard examples: a set exists in virtue of its members, a conjunction is true in virtue of both of its parts, and a statue has its shape in virtue of the arrangement of its clay.

On most accounts, grounding has these features:

- **Irreflexive**. Nothing grounds itself.
- **Asymmetric**. If A grounds B, B does not ground A.
- **Transitive**. If A grounds B and B grounds C, then A grounds C.
- **Necessitating**. A full ground guarantees what it grounds.
- **Downward in fundamentality**. The ground is more fundamental than what it grounds.

Kit Fine, Jonathan Schaffer and Gideon Rosen developed the theory. Jessica Wilson argued that “grounding” lumps together many specific relations and should be replaced by them.

This distinction is central to the thesis. The infographic says ground is “not an earlier event in time” and “not a cause inside the order.” That makes it sound like metaphysical grounding. But a classical creator is usually thought of as a *productive* source, and a freely chosen creation does not *necessitate* its result. Chapter 9 shows that God* needs a broader notion, written **ground***.

Ladder — grounding

Rung 1. Some “because”s are about what happened first: the tower fell because I kicked it. Other “because”s are about what something *is*: the tower is tall because it has lots of blocks. The second kind has no “first.”

Rung 2. Causation links events in time. Grounding links facts by what makes them so: “this shape is a square in virtue of having four equal sides and four right angles.” A ground doesn’t come before in time. It is what the fact holds in virtue of.

Rung 3. Grounding is usually treated as a strict partial order: irreflexive, asymmetric and transitive. It is also hyperintensional. It separates necessarily equivalent facts: {Socrates} exists in virtue of Socrates, and not the other way round, though each exists exactly when the other does. Full grounds necessitate what they ground. Grounding backs a distinctive kind of *metaphysical explanation*.

Rung 4. Open issues include:

- Is grounding one relation (Fine, Schaffer) or a family (Wilson’s “small-g” relations)?
- Is it well-founded?
- Does it necessitate? Contingent grounding has its defenders.
- Is it a relation between facts (Fine) or between entities (Schaffer)?
- How does it relate to causation? Schaffer (2016) models grounding on structural equations.

For God*, the necessitation question decides whether a free creator counts as a ground. Chapter 9 answers by going generic.

2.5 Necessity and contingency

A truth is **necessary** if it could not have been false, and **contingent** if it could have been. Philosophers separate grades of necessity:

- **logical**: “all bachelors are unmarried”;
- **metaphysical**: “water is H₂O,” according to Kripke;
- **nomic or physical**: “nothing travels faster than light,” given our laws.

A useful picture is *possible worlds*: complete ways reality could have been. Necessary truths hold in all of them, contingent truths in some.

Two points matter for God*. First, the physical world appears contingent: it seems it could have failed to exist, or had different laws. Many arguments for a ground start there. Second, some naturalists reply that the physical world, or its first state, is *necessary*, so nothing outside it is needed. The infographic's constraint that "necessity given the laws does not fill the role" handles the nomic grade. Chapter 10 extends it to metaphysical necessity.

Ladder — necessity

Rung 1. Some things couldn't be any other way: a triangle always has three sides. Other things just happen to be true: your shirt is blue today.

Rung 2. Philosophers imagine "possible worlds," every complete way things could have gone. Necessary means true in every one. Contingent means true in some and not others.

Rung 3. The grades differ in which worlds count: all logically possible worlds, all metaphysically possible ones, or only those with our laws. A *necessary being* exists in every metaphysically possible world. The cosmological argument moves from the contingency of the world to a necessary being.

Rung 4. Deep questions:

- Is metaphysical necessity grounded in essence (Fine 1994)?
- Can a contingent world be explained by a necessary being without the world becoming necessary too? This is the *modal collapse* problem (§2.6).
- Is a "necessary initial state" of the universe coherent (Oppy)?

These decide whether "the universe exists necessarily" counts as an outside account or as Empty. This thesis argues for Empty (§10.4).

2.6 Brute facts and the principle of sufficient reason

A **brute fact** is a fact with no explanation at all. The **principle of sufficient reason** (PSR), stated most famously by Leibniz (→ A.10), denies that there are any: every fact has a reason why it is so and not otherwise.

The PSR is the engine of most arguments that God* is Filled. If every fact has an explanation, so does the fact that the physical world exists. An infinite chain of physical causes explains each member but not the whole chain. So the explanation must lie outside the chain.

The most famous objection is **modal collapse** (van Inwagen 1983; Bennett's earlier version). Take the Big Conjunctive Contingent Fact: all contingent truths joined into one. By the PSR it has an explanation. A contingent explainer would be part of what it explains. A necessary explainer that *entails* it would make it necessary. So the PSR seems to make everything necessary. Defenders reply that an explanation need not entail what it explains. A free choice, or a probabilistic cause, can explain without necessitating (Pruss 2006).

This debate sits directly under the verdicts. A reader who rejects the PSR can accept a brute physical world, which is **Empty**. A reader who holds the strong PSR must find a non-internal ground (**Filled** or **Split**), make the world necessary (Empty, by §10.4), or reject the question (**Refused**).

Ladder — the PSR and modal collapse

Rung 1. The rule “everything has a reason” sounds right. But if everything has a reason, then “everything that happens” needs a reason too. That reason can't be one of the things that happens, or it would be explaining itself.

Rung 2. Leibniz said the world as a whole must have a reason outside it: a being that exists necessarily. Critics say that if that reason *forces* the world to be as it is, nothing could have been different, and that seems false.

Rung 3. Let p be the conjunction of all contingent truths. If the PSR holds, some q explains p . If q is contingent, it is a conjunct of p , so the explanation is circular. If q is necessary and q entails p , then p is necessary, and there are no contingent truths. Responses: deny that explanation requires entailment, restrict the PSR to positive facts or existence facts, or accept a brute fact.

Rung 4. Pruss defends a PSR restricted to contingent facts, with libertarian free choice as an explanation that does not necessitate. Della Rocca defends a thoroughgoing PSR and accepts its monistic, Spinozist consequences. Dasgupta's work on *autonomous* facts suggests some facts are neither grounded nor brute. God* treats these as rival paths to different verdicts. It takes no side on which is correct.

2.7 Wholes, parts and the fallacy of composition

The definition speaks of the physical order “as a whole.” Two warnings belong here.

The fallacy of composition. What is true of each part need not be true of the whole. Each brick is small, but the wall is not. In a 1948 radio debate, Bertrand Russell told Frederick Copleston that every human has a mother, but the human race does not. By analogy, each thing in the universe may have a cause without the universe having one (→ A.19).

Hume's twenty particles. Hume argued that if you have explained each of twenty particles of matter, it would be unreasonable to ask for a further explanation of the twenty together. Grouping them is “an arbitrary act of the mind” (→ A.12).

These warnings do not make the question illegitimate. They show that asking about the whole is a *separate* question from asking about the parts, which is exactly what the God* definition claims. Chapter 13 shows that the question needs no special “whole” object at all. Modern *plural logic* can talk about “the physical things, taken together” without assuming a single object that contains them.

2.8 Presupposition

“Have you stopped cheating at cards?” cannot be answered well with “yes” or “no” if you never cheated. The question *presupposes* something false. P. F. Strawson (1950) argued that sentences with failed presuppositions, like “the King of France is bald,” are neither true nor false. The right response is to reject the question. The **Refused** verdict is this move made about God*: not “no one fills the role” but “the role question presupposes something false.” Chapter 11 names the three presuppositions at stake.

Chapter 3 — A short history of the question of an ultimate ground

People in every major civilization have asked why there is a world and why it is orderly. They have also, again and again, separated those two questions without naming the separation. This chapter tells that history in nine stops. Its thesis is simple: **the split between G-existence and G-law is old and recurrent, and the God* framework names a distinction that the history already contains.** Figure 3.1 lays out the eras by tradition. Appendix A gives a primer on each figure.

The ground question was asked in every tradition, in every era

Key figures by tradition and era; columns are eras, not to scale

	Before 1 CE	1-1500	1500-1900	1900-today
Greek and European	Thales, Heraclitus Parmenides, atomists Plato, Aristotle the Stoics	Plotinus (c. 250) Augustine (c. 400) Aquinas (c. 1270)	Descartes, Newton Spinoza, Leibniz Hume, Kant, Hegel	Whitehead, Tillich Carnap, Russell Lewis, Fine, Oppy
Abrahamic	Genesis 1 2 Maccabees 7:28	Irenaeus, Theophilus Avicenna, al-Ghazālī Maimonides	Boyle, Clarke the deists	Plantinga, Swinburne Craig, Pruss
Indian	Rig Veda 10.129 the Upaniṣads the Buddha, Mahāvīra	Nāgārjuna (c. 150) Śaṅkara (c. 700) Udāyana, Rāmānuja Madhva, Guru Nānak	the Sikh Gurus Navya-Nyāya	global philosophy of religion
Chinese	Daodejing Zhuangzi	Guo Xiang (d. 312) Zhu Xi (1130-1200)	Korean li-qi debate (Yi Hwang, Yi I)	Needham (1951) comparative study

Figure 3.1 · Three thousand years of the ultimate-ground question

3.1 Greek beginnings: stuff and pattern

The first Greek philosophers, in Ionia in the sixth century BCE, asked for the *archē*, the origin or principle of all things.

- **Thales** said water.
- **Anaximander** said the *apeiron*, the boundless.
- **Heraclitus** said the world is ever-changing fire, but that its changes follow a *logos*, a rational measure or law.

So already there are two kinds of answer: a *stuff* the world is made of, and a *pattern* the world follows. These are ancestors of G-existence and G-law (→ A.1).

Parmenides added the principle later summed up as *ex nihilo nihil fit*: nothing comes from nothing. Almost every later thinker had to respond to it. Some concluded that the world is eternal. Others concluded that a world with a beginning needs a source that is not itself part of the world.

The atomists — Leucippus, Democritus and later Epicurus and Lucretius — gave the first fully internal account: atoms and void, eternal, with no designer. The Greek gods existed, but inside the cosmos and made of atoms like everything else. In the framework's terms that is **Empty with gods**, a useful reminder that believing in gods and filling God* are different things.

3.2 Plato: a craftsman and a pattern

Plato's *Timaeus* (→ A.3) tells a "likely story." A divine craftsman, the **Demiurge**, looks to the eternal **Forms** as a model and brings order to a chaotic, pre-existing receptacle. The craftsman does not create the raw material. The Forms are not made by the craftsman. In the *Republic*, the Form of the Good is "beyond being" and is the source of the other Forms' being and intelligibility.

Plato therefore offers at least two candidates for grounding order (craftsman and Forms) and arguably none for the bare existence of the receptacle. Read that way, this is the infographic's **Split**. Another reading makes the Good the single source of everything, which would be **Filled**. The framework does not say which reading is right. It makes the choice between them visible.

3.3 Aristotle: an eternal world and an unmoved mover

Aristotle (→ A.4) held that the cosmos has always existed. Motion needs explaining, and an infinite chain of movers explains nothing as a whole. So he posited an **unmoved mover**: pure actuality, thought thinking itself. It moves the heavens not by pushing but as an object of love and aspiration, a *final* cause. It does not create matter or explain why there is a cosmos rather than none.

Aristotle is the clearest ancient case of a ground of *order and motion* with no ground of *existence*: **Split** (law-side only).

3.4 Stoics and Neoplatonists: inside and beyond

The Stoics identified God with the *logos*, a fiery, rational, material principle running through the cosmos. God is the world's mind, but made of the world's stuff. By the infographic's rule, "a highest being inside the cosmos does not count," so Stoicism is **Empty (immanent deity)**. This result is surprising and useful: God* does not track whether a view is religious.

Plotinus (third century CE, → A.5) went the other way. Everything flows, or "emanates," from **the One**, a principle so simple it is beyond being and beyond description. From the One comes Intellect (the Forms), then Soul, then the physical world. The One is outside

the physical order and is the source of both its existence and its intelligible structure. That is **Filled**, with an impersonal occupant that emanates rather than chooses.

3.5 The Abrahamic traditions: creation from nothing

The Hebrew Bible opens with God creating “the heavens and the earth.” Scholars debate whether Genesis 1 describes creation from nothing or the ordering of a watery chaos. The explicit doctrine of **creation ex nihilo** is stated in 2 Maccabees 7:28 and was developed by Christian writers of the second century, such as Theophilus of Antioch and Irenaeus, partly against Platonist views of eternal matter (May 1994). From then on, Judaism, Christianity and Islam taught that one God is the source of *both* the world’s existence and its order.

Four ideas from this tradition shape the God* definition.

- **Augustine** (→ A.6) argued that God made time along with the world. There was no “before” in which God waited. A ground is not an earlier event.
- **Avicenna** (Ibn Sīnā, → A.7) distinguished a thing’s *essence* (what it is) from its *existence* (that it is). Everything whose essence does not include existence is merely possible in itself and needs a cause of its existence. The chain ends in the **Necessary Existent**, whose essence is to exist. This is the clearest historical source for the G-existence side.
- **Al-Ghazālī** (→ A.7) argued that the regularities of nature are God’s habitual acts, not necessary connections. That makes God the ground of *law* as directly as of existence. **Maimonides** (→ A.7) held that creation cannot be demonstrated by reason but is better supported than Aristotle’s eternal world.
- **Thomas Aquinas** (→ A.8) gave the Five Ways. He also argued, in *On the Eternity of the World*, that reason alone cannot prove the world had a beginning, and that a world with no beginning would *still* depend on God for its existence at every moment. This is the historical root of the infographic’s constraint: “Ground is not an earlier event in time.”

The Sikh *Mūl Mantar* opens the Guru Granth Sahib with *Ik Onkar*: one creator, beyond time, self-existent. It fits the same pattern (→ A.9).

3.6 India: order, emptiness and many schools

Indian philosophy offers the widest range of positions on the ultimate-ground question in any single civilization (→ A.9).

- **The Vedas.** They speak of *ṛta*, the cosmic order that keeps the sun on its course and truth in speech: an early concept of G-law. The *Nāsadīya Sūkta* (Rig Veda 10.129) asks how creation arose and ends, “he who surveys it in the highest heaven, he surely knows — or perhaps he does not know.” It is the most famous ancient statement that the question may outrun knowledge. That is an epistemic state, not a verdict.
- **The Upaniṣads.** They identify **Brahman** as the ground from which beings arise, by which they live, and into which they return.
- **Classical Sāṃkhya.** It explains the world as the unfolding of *prakṛti* (primal matter) in the presence of *puruṣa* (pure consciousness), with no creator. The infographic maps it to **Empty**.
- **Nyāya-Vaiśeṣika.** Udāyana, around 1000 CE, gave careful arguments for **Īśvara**, a Lord who arranges eternal atoms. Eternal atoms are not created, so this is plausibly **Split** (law-side) rather than Filled.
- **Mīmāṃsā, Jainism and most Buddhist schools.** They argue *against* a creator. The universe is beginningless and runs by its own laws, such as karma.
- **Madhyamaka Buddhism.** Nāgārjuna, around 150 CE, argued that nothing has *svabhāva*, independent intrinsic existence. So there is no self-standing “whole” to be grounded, and no self-standing ground. This is the paradigm of **Refused**.
- **Vedānta.** It split into rival schools on how Brahman relates to the world. Śaṅkara’s *Advaita* holds that Brahman alone is ultimately real and the world is an appearance. Rāmānuja’s *Viśiṣṭādvaita* holds that the world is Brahman’s body. Madhva’s *Dvaita* holds that Brahman is a distinct creator. Chapter 10 scores Advaita.

3.7 China: the Dao, principle and material force

The Daodejing (→ A.9) opens by saying that the Dao that can be spoken is not the constant Dao. Chapter 42 says “the Dao gives birth to one, one gives birth to two, two to three, three to the ten thousand things.” Read as a cosmogony, the Dao is a ground of both the existence and the patterning of things: **Filled**, impersonal. Read as the way things themselves go, with no source outside them, the Dao is not a ground of a whole at all: **Refused** or **Empty**. The infographic records the refusing reading.

Neo-Confucianism gives the most exact historical parallel to the God* distinction. **Zhu Xi** (1130–1200) analysed every thing into *qi*, the material force that makes it concrete, and *li*, the principle or pattern that makes it what it is. He held that li and qi are never apart, but that li is “prior” in a logical sense. All li are one in the **Supreme Ultimate** (*taiji*). Qi

corresponds to G-existence and li to G-law. The Supreme Ultimate is a candidate ground of the law-side, and the debate over whether it also grounds qi is exactly the question “Filled or Split?”

The historian of science Joseph Needham (1951) argued that Chinese thought pictured the world's order as an organic harmony rather than a code issued by a lawgiver. That is a reminder that “law” itself is a metaphor, which Chapter 4 examines.

3.8 Early modern Europe: laws, clocks and the PSR

The scientific revolution made laws of nature central, and with them the G-law side.

- **Descartes** derived the laws of motion from God's unchanging nature (*Principles* II.36–37).
- **Boyle and Newton** spoke of laws as God's decrees. Newton's *General Scholium* (1713) inferred an intelligent lawgiver from the solar system's order.
- **The Leibniz–Clarke correspondence** (1715–16) mocked Newton's God as a clockmaker who must keep winding his clock. The dispute was over *how* God grounds law: once, perfectly, or continually.
- **Deism** (→ A.11) pictured God as a starter who creates the world and its laws, then leaves them to run. The infographic lists “a starter after which law needs no further ground” under Split. Chapter 10 revisits that placement.
- **Spinoza** (1677, → A.10) identified God with Nature, *Deus sive Natura*, the one substance of which everything is a mode.
- **Leibniz** (→ A.10) stated the PSR sharply and asked the question in its most famous form: why is there something rather than nothing?
- **Hume's *Dialogues concerning Natural Religion*** (1779, → A.12) attacked both the design argument and the cosmological argument.
- **Kant** (1781, → A.13) argued that the cosmological argument secretly depends on the ontological argument, and that reason falls into contradictions (“antinomies”) when it asks whether the world has a beginning.

3.9 From the nineteenth century to today

The modern story has four threads.

1. **Critique and retreat.** Laplace reportedly told Napoleon he had “no need of that hypothesis,” meaning God as a mechanic who adjusts orbits. Darwin removed the need for a designer of organisms, though not of the laws. The logical positivists declared the question meaningless (→ A.21).

2. **Theology of ground.** **Paul Tillich** (→ A.14) said God is not a being, not even the highest, but *being-itself*, the “ground of being.” **Martin Heidegger** made “Why are there beings at all instead of nothing?” the first question of metaphysics. **Whitehead’s process philosophy** (→ A.15) made God one factor among others: God supplies the world’s ordering possibilities but not its raw creativity, which yields **Split**.
3. **Revival.** From the 1960s, analytic philosophers such as Alvin Plantinga, Richard Swinburne and William Lane Craig rebuilt the arguments of natural theology with modern logic. Naturalists such as Graham Oppy answered with equal rigour (→ A.16). The cosmological argument came back in several forms, including Craig’s *kalām*, Pruss’s Leibnizian version and O’Connor’s necessary-being version.
4. **New tools.** The *grounding* revolution of the 2000s supplied a vocabulary for non-causal dependence (§2.4). Physical cosmology, from the Big Bang to inflation and quantum cosmology, supplied real data about the early universe (Chapter 5). *Global* or *comparative* philosophy of religion began deliberately testing its categories on non-Western traditions (Chapter 8).

3.10 What the history teaches

The table below previews the pattern. Many thinkers ground one side of the physical order but not the other. The full verdicts, with reasons, come in Chapter 14.

Thinker or school	Ground of existence?	Ground of order or law?	Preview verdict
Atomists	None: atoms are eternal	None: necessity and chance	Empty
Plato (<i>Timaeus</i>)	None for the receptacle	Demiurge and Forms	Split
Aristotle	None: the world is eternal	Unmoved mover, as final cause	Split
Stoics	Divine fire, inside the world	Logos, inside the world	Empty (immanent deity)
Plotinus	The One	The One, via Intellect	Filled
Avicenna, Aquinas	Necessary Existent, God	God	Filled

Thinker or school	Ground of existence?	Ground of order or law?	Preview verdict
Nyāya (Udāyana)	None: atoms are eternal	Īśvara arranges	Split
Zhu Xi	Qi: debated	Li, the Supreme Ultimate	Filled or Split, by reading
Deism	God	God, once only	Filled (§10.9)
Whitehead	Creativity, not God	God's primordial nature	Split
Nāgārjuna	The question misfires	The question misfires	Refused

Key idea. The question “what grounds existence?” and the question “what grounds order?” have separate histories. Many great systems answer one and not the other. A shared vocabulary that cannot say so is too blunt.

Chapter 4 — Laws of nature: what G-law is about

G-law is the fact that the physical domain is “lawful — regular and structured, not a heap of disconnected happenings.” What counts as grounding that fact depends on what laws of nature *are*. Philosophers disagree deeply here. This chapter lays out the options, because each theory of laws changes which God* verdicts are open.

4.1 Laws versus accidents

“All copper conducts electricity” is a law. “All the coins in my pocket are copper” is an accident. Both are true generalizations. The difference shows in **counterfactuals**. If this wire were copper, it *would* conduct. If this coin were in my pocket, it would *not* thereby become copper. A classic pair makes the point.

- “No sphere of uranium-235 is a mile wide” is a law-like truth, because such a sphere would explode long before reaching that size.
- “No sphere of gold is a mile wide” is true only by accident.

Any theory of laws must explain this difference.

4.2 Four main theories, and a fifth

1. **Humean regularity, or the “best system” (Lewis).** There is just the “mosaic” of particular facts: what is where, when. The laws are the axioms of the best summary of the mosaic, the one that best balances simplicity and strength. Laws describe; they do not push.
2. **Governing laws (Armstrong, Dretske, Tooley).** A law is a relation of *necessitation* between properties, or universals: being F necessitates being G. Laws are something over and above the mosaic, and they explain it.
3. **Dispositional essentialism (Bird; Ellis; Mumford).** Properties are *powers*. Having negative charge just *is* being disposed to repel other negative charges. Laws flow from the essences of properties and are metaphysically necessary given those properties.
4. **Primitivism (Maudlin).** Laws are fundamental and cannot be reduced to anything else. They produce the evolution of the world’s states.
5. **Theistic accounts.** The regularities are God’s decrees (Descartes, Boyle; defended recently by Foster 2004), or God’s habitual actions with no necessary connections in nature (al-Ghazālī’s occasionalism).

Figure 4.1 places these on two axes. Do laws merely *summarize*, or do they *do work*? Is their source *inside* nature or *beyond* it?

Theories of laws differ on what laws do and where they come from

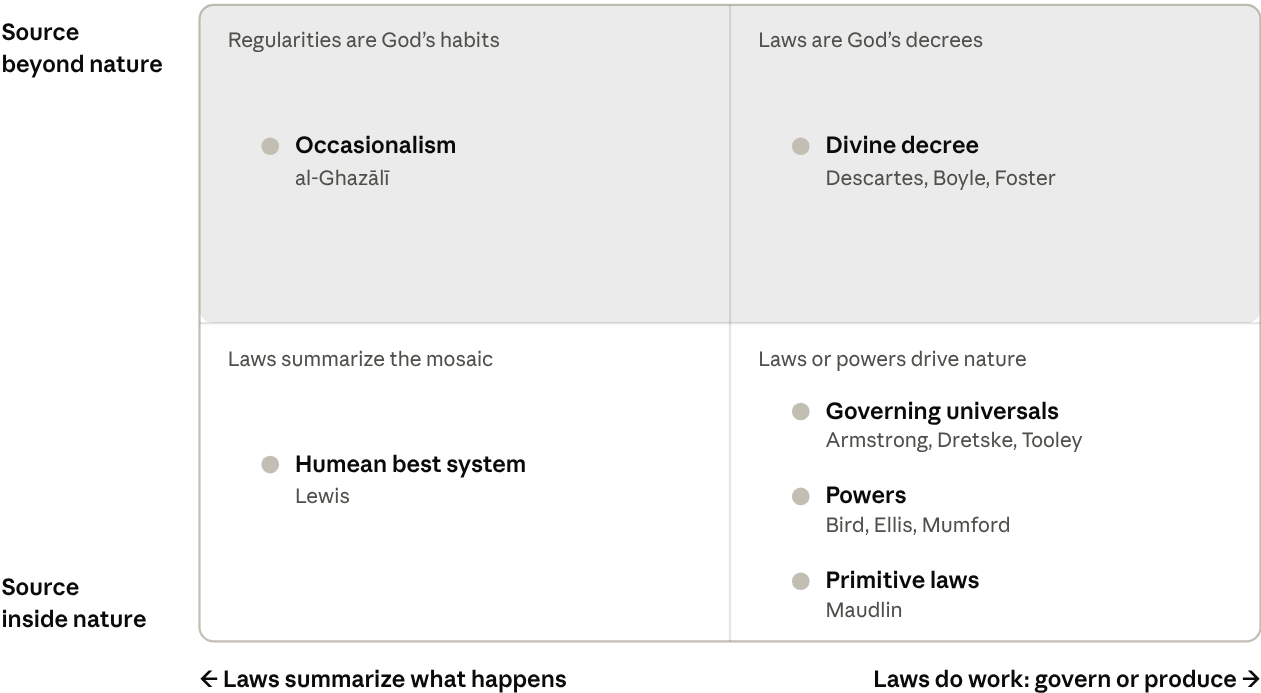


Figure 4.1 · Four theories of what a law of nature is, plus two theistic ones

The shaded upper half contains the theistic accounts, on which lawfulness is grounded from beyond nature. They answer G-law by construction. The lower half leaves G-law open.

Ladder — what is a law of nature?

Rung 1. Things in nature always behave in the same ways: dropped things fall, ice melts when warm. A “law of nature” is a name for one of those always-the-same ways.

Rung 2. Scientists state laws as rules, like $F = ma$. Philosophers ask whether such a rule is only a short description of what always happens, or something that *makes* things happen that way. A traffic law does make people drive on one side of the road. Is a law of nature more like a traffic law or more like a weather report?

Rung 3. Humeans take the second view: laws supervene on the mosaic, and the best-system account explains why some regularities count as laws. Anti-Humeans, including governing, powers and primitivist views, say laws or powers are ontologically prior and *explain* the regularities. The main objection to Humeanism: if laws only describe, the universe’s pervasive regularity looks like a colossal coincidence.

Rung 4. Current work covers:

- Humean accounts that are sensitive to the best system for *us* (Hicks, Jaag and Loew; Dorst).
- Minimal primitivism (Chen and Goldstein).
- The inference problem for governing views: how does $N(F,G)$ *entail* that all Fs are G? (van Fraassen 1989)
- Whether dispositional essentialism can accommodate conservation laws and symmetries.

For God*, what matters is whether lawfulness is a *further* fact beyond existence (anti-Humean), or a structural feature of the existing mosaic (Humean). Section 4.3 draws the consequences.

4.3 How the theory of laws changes the verdicts

This is the chapter’s main result. It is also a new observation about the infographic: **G-law’s independence from G-existence is not guaranteed; it depends on the theory of laws.**

Theory of laws	Is lawfulness a fact beyond existence?	Consequence for the God* verdicts
Humean best system	Only in part. Lawfulness is the mosaic's compressibility.	A ground of existence that fixes the mosaic fixes the laws too. Split can remain only if existence is brute and its orderliness has a separate ground.
Governing (Armstrong)	Yes. Necessitation relations are extra entities.	All four verdicts are open. Split is natural: the matter and the universals can have different sources.
Dispositional essentialism	No. Laws follow from which properties exist.	A ground of the existence of propertied things automatically grounds the laws. "Law-ground without existence-ground" is still possible, but mainly as Split in the style of Plato and Aristotle, where an orderer works on pre-existing stuff.
Primitivism	Yes. Laws are basic.	All four verdicts open.
Divine decree or occasionalism	Yes, and it is grounded.	By construction, Filled or Split.

Key idea. The two sides of the arch are two questions only if lawfulness is something over and above existence. God* should say so, and it should let each theory of laws decide whether its Split cases are live. This thesis adopts that clarification in God* v2 (Chapter 16).

4.4 “Why these laws?” and fine-tuning

Beyond “why is there law at all?” lies “why *these* laws and constants?” Several constants of physics appear to lie in narrow ranges that permit complex chemistry and life. This is the **fine-tuning** observation. Its standard explanations are:

- design by an agent;

- a multiverse with many values, plus the selection effect that observers can only find themselves where life is possible;
- a deeper law that fixes the values;
- brute luck.

Fine-tuning is evidence about the *occupant* of God*: whether it has preferences, for instance. It is not part of the role. It also shows that G-law has a *specific* component (which laws) as well as a *generic* one (that there are laws). Chapter 13 folds the specific component into G-law.

Chapter 5 — What physics can and cannot settle

The infographic says the Big Bang, inflation, bounce cosmologies, quantum gravity, dark energy and multiverse generators are “internal unless they can be stated without assuming a physical order.” This chapter explains each and shows why the rule is sound.

5.1 The Big Bang

In 1927 Georges Lemaître, a Belgian physicist and priest, showed that Einstein’s equations allow an expanding universe. Edwin Hubble’s galaxy observations (1929) supported expansion. Arno Penzias and Robert Wilson’s discovery of the cosmic microwave background (1965) confirmed that the early universe was hot and dense. The Big Bang theory says that the observable universe was in that state about 13.8 billion years ago and has expanded and cooled since.

The theory does **not** say that the universe came from nothing. Run the equations backward and density grows without limit. The **singularity**, the “ $t = 0$ ” where the equations break down, marks where classical theory stops being reliable, not a proven first moment. Lemaître himself warned Pope Pius XII in 1951 against treating the Big Bang as proof of creation.

5.2 Singularity theorems and the BGV theorem

Hawking and Penrose proved in the 1960s and 1970s that, under general conditions, classical general relativity predicts singularities. In 2003, Borde, Guth and Vilenkin (BGV) proved that any spacetime that has on average been expanding throughout its history is *past-incomplete*: its past cannot be extended forever. Some philosophers read this as evidence of an absolute beginning. The authors and most cosmologists read it more narrowly. The theorem shows that classical expanding spacetime had a past boundary. It does not show what, if anything, lies beyond that boundary in a quantum description.

5.3 Inflation and the multiverse

Alan Guth (1981) proposed **inflation**: a brief period of exponential expansion in the very early universe. It explains why the universe is so uniform and flat. In many versions, inflation is *eternal*. It stops in some regions, which become “bubble universes,” while continuing elsewhere. Combined with string theory’s many possible vacuum states, this yields a **multiverse** in which constants vary between bubbles.

For God*, the key point is that the multiverse generator is itself a physical process, run by an inflaton field under quantum laws. It explains why there are *many* universes, given a lawful physical order. It does not explain why there is a lawful physical order. That is why the infographic calls it internal, and why Chapter 12 extends “the physical order” to include the whole multiverse.

5.4 Quantum cosmology and “something from nothing”

Two famous proposals describe a universe with no classical beginning.

- **Alexander Vilenkin (1982)** described a universe “tunnelling from nothing.”
- **James Hartle and Stephen Hawking (1983)** proposed the *no-boundary* wave function, in which time becomes space-like near the beginning, so there is no edge. Asking what came before is like asking what lies north of the North Pole.

In 2012 Lawrence Krauss’s book *A Universe from Nothing* presented such physics as answering why there is something rather than nothing. The physicist-philosopher David Albert replied in *The New York Times*. The “nothing” in these models is a *quantum-field vacuum* or a *state with no classical spacetime*, still governed by the laws of quantum mechanics. Physics explains how one physical state arises from another, given laws. It cannot explain, by those laws, why there are laws and states at all.

That is the infographic’s internal/non-internal rule in one example.

Ladder — can physics get a universe from nothing?

Rung 1. Scientists sometimes say a universe can pop out of “nothing.” But their “nothing” still has rules for how things can pop. Real nothing wouldn’t even have rules.

Rung 2. A vacuum in physics is not empty. It is the lowest-energy state of fields, and it bubbles with activity. A universe arising from a vacuum is a change from one physical situation to another. It is a real discovery, but it answers “how did our universe start?”, not “why is there anything at all?”

Rung 3. Vilenkin’s tunnelling and Hartle–Hawking’s no-boundary proposal assign amplitudes to three-geometries, using a wave function that obeys the Wheeler–DeWitt equation. Both presuppose a Hilbert space, a Hamiltonian constraint and quantum laws. The explanandum “there is a lawful physical order” is assumed, not derived. Albert (2012) makes exactly this point against Krauss.

Rung 4. A more sophisticated reply asks whether a *self-explanatory* physics is possible, for example one in which the laws are the unique consistent possibilities, as in Wheeler’s “law without law” or some structural-realist programmes. If such a physics were metaphysically necessary, it would ground E and L internally, as a necessary cosmos. God* scores that **Empty**, not Filled, since nothing non-internal is invoked (§10.4). The framework thus separates *physics-as-occupant*, which is impossible because physics is internal, from *physics-as-making-the-role-idle*, which is possible and is one way to reach Empty.

5.5 Bounces, cycles and a past-eternal universe

Some models replace the Big Bang with a **bounce** from a previous contracting phase. Others propose endless **cycles**: Steinhardt and Turok’s ekpyrotic model, and Penrose’s conformal cyclic cosmology (2010). If any is right, the physical order may have no first moment. Many people assume that would remove the need for a ground.

The history in Chapter 3 shows why that does not follow. Aquinas and Avicenna both held that a beginningless world would still need a ground of its existence. Aristotle held that an eternal world still needs a ground of its motion. A past-eternal cosmos changes *which occupant theories* are plausible. It does not answer the role question. The infographic’s phrase “a beginningless or brute physical whole still sits here” in the Empty panel is

correct, but incomplete. Someone who says “the cosmos is beginningless, and that is the end of it” is giving Empty. Someone who says “the cosmos is beginningless *and* sustained by God” is giving Filled.

5.6 The general principle

Every explanation in physics has the form **later or other physical states = laws applied to earlier or other physical states**. Figure 5.1 draws this. Every arrow physics can draw runs between things inside the box. The God* question is about the box itself.

Physics draws arrows inside the box; the role asks about the box

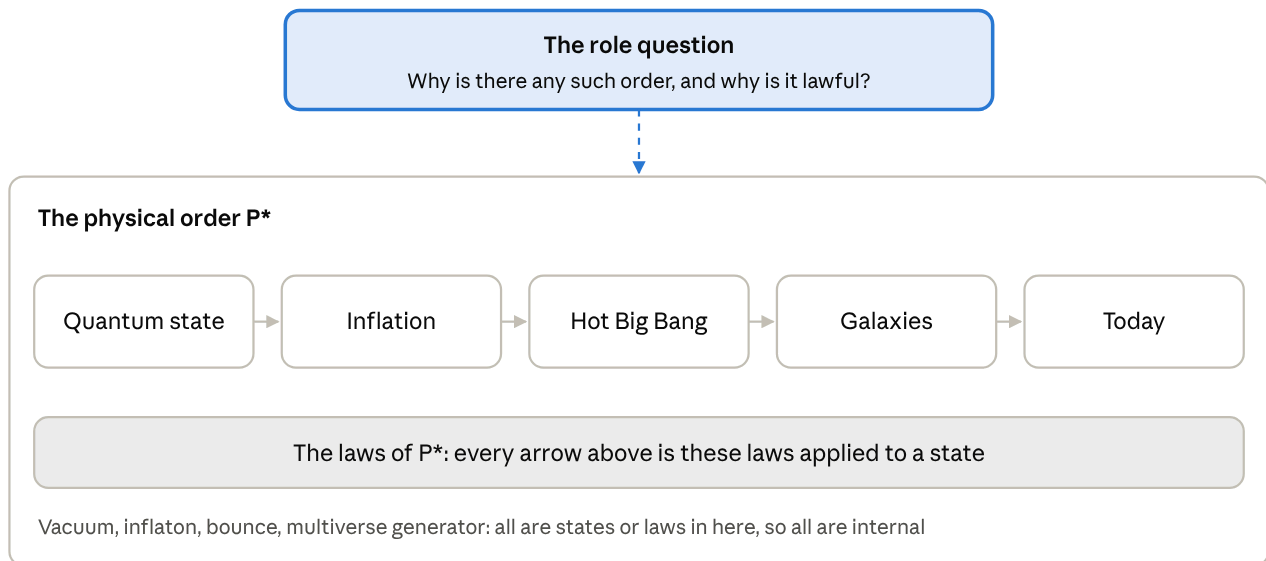


Figure 5.1 · Every physical explanation starts from something physical

This does not make physics irrelevant to the God* question. Physics can discover:

- whether the physical order had a first state;
- whether its laws are unique or fine-tuned;
- whether there are many universes.

Such discoveries bear on which occupant theories are plausible, and so on which verdict is more likely. What physics cannot do, by its own methods, is occupy the role.

Key idea. “Internal” is not a criticism of physics. It is a statement of physics’ job description. Physics explains the physical order from the inside. The role asks about it from outside, and some traditions deny that there is an outside question at all.

Part II — The God* framework

Chapter 6 — The definition, made precise

6.1 The definition as given

The infographic states:

God* is the title of a role, not the name of a being. The role is occupied by whatever, if anything, is the non-internal ground of the physical order as a whole. That whole has two sides. **G-existence**: that there is a physically structured domain at all, rather than no such domain. **G-law**: that this domain is lawful — regular and structured, not a heap of disconnected happenings. God* names one non-internal ground of both.

Every phrase is doing work. The table below parses them and says where each is examined.

Phrase	What it does	Examined in
“title of a role, not the name of a being”	Makes God* a Ramsey–Lewis role term, so “no occupant” is a coherent answer	§2.1–2.2
“whatever, if anything”	Leaves the occupant’s kind open (being, principle, value, structure) and allows none	§10.6
“non-internal”	Excludes anything whose statement already assumes a physical order	§12.3
“ground”	The dependence relation between occupant and order	Chapter 9
“the physical order”	The domain to be grounded	Chapter 12
“as a whole”	The whole, not its parts one by one	§13.1
G-existence, G-law	The two sides, or explananda	Chapter 4, §13.3
“one ... of both”	Unity: a single occupant for both sides	§7.2, §13.4

6.2 A formal statement

Readers new to logic can skip this section. The symbols are a compact restatement of 6.1.

The notation:

- **P** is the physical order: the physical things, taken together, with their relations.
- **E(P)** is the fact that there are physical things (G-existence).
- **L(P)** is the fact that they are lawful, including which laws hold (G-law).
- **Int(x)** says x is internal to P.
- **G*(x, f)** says x grounds* fact f.

The God* role is the open formula:

$$R(x) \equiv \neg \mathrm{Int}(x) \vee G^*(x, E(P)) \vee G^*(x, L(P))$$

and the title is the definite description:

$$\mathrm{God}^* \equiv \iota x, R(x) \quad \text{(defined only if } \exists x! x, R(x) \text{)}$$

The **Ramsey sentence** $\exists x R(x)$ says the role is played. Lewis's uniqueness clause says the title *refers* only if exactly one thing plays it. A second, weaker formula matters for the Split and Empty verdicts:

$$R^-(x) \equiv \neg \mathrm{Int}(x) \vee \big(G^*(x, E(P)) \vee G^*(x, L(P))\big)$$

$R^-(x)$ says that x is a non-internal ground of *at least one* side. Every Filled occupant satisfies R^- , but not the other way round.

6.3 What the role contains, and what it leaves out

The God* role is deliberately thin. It says nothing about goodness, mind, will or worship. The infographic puts this as “Occupant theories are not the role.”

In the role	Occupant theory, not in the role
Not internal to the physical order	Personal or impersonal
Grounds that there is a physical order	Good, loving, just

In the role	Occupant theory, not in the role
Grounds that it is lawful, and which laws hold	Omnipotent, omniscient
One and the same ground for both	Worthy of worship
	Revealed in scripture, prophets, avatars
	Trinity, covenant, Qur'an, incarnation
	Necessary or contingent in its own existence
	Acts once or sustains continually

Why so thin? A thin role is what makes the vocabulary *shared*. Think of cholera. In 1854 John Snow could argue about “the cause of cholera” with miasma theorists before anyone knew it was the bacterium *Vibrio cholerae*. The shared term named a job. The disagreement was over who did it. God* does the same for the ground of the physical order.

The cost of thinness. Three consequences follow, and they are features rather than bugs.

1. **Classical theists will say God is *more* than God*.** Filled is necessary for their view, not sufficient. Their God is the occupant *plus* the occupant theory.
2. **Someone can believe in gods while God* is Empty.** Examples are the Stoics, Epicurus, the Homeric gods, and any deity that arises inside the cosmos. The framework exposes this real disagreement, which the word “God” hides.
3. **Someone can deny “God” while God* is Filled.** An axiarchist, who holds that goodness itself explains why the world exists, may reject every personal god yet fill the role (§10.6).

Key idea. God* is the minimal “foundational” question. It is compatible with any answer about the character of the foundation.

6.4 The keystone metaphor

The infographic draws the role as an arch. G-existence and G-law are the two springers, the stones at each foot. The physical order is the space beneath. God* is the keystone slot at the top: “one stone that locks both sides — occupied, or not.” Figure 6.1 redraws it with this thesis’s labels.

Figure 6.1 · The keystone: God* as a role over two sides (after the original in...

The metaphor fits better than it may first appear, in two ways.

- **A keystone is a *structural* ground, not an earlier event.** It holds the arch *now*, not by having been placed first. In fact it is laid *last*. That matches Aquinas’s point that dependence does not require a beginning.
- **Some arches stand without a keystone.** A *corbelled* arch, like the one in the Mycenaean “Treasury of Atreus,” is made of stones that each overhang the one below until they meet. Each stone is held by the stones beneath it, and there is no locking stone at the top. That is a picture of **Empty**: an order that stands with nothing at the apex. A future version of the infographic could draw the Empty panel as a corbelled arch rather than an arch with an empty slot. It would show Empty as a positive structural thesis, not a gap.

Chapter 7 — Four verdicts and a decision procedure

7.1 Three presuppositions

The role question “what fills R(x)?” presupposes three things. Chapter 11 defends this list.

- **π_1 , totality.** There are physical things, and it makes sense to talk about all of them together.
- **π_2 , explicability.** It makes sense to ask what that totality exists and is lawful *in virtue of*. Dependence relations can in principle apply to the totality, not only within it.
- **π_3 , determinacy.** Either something is a non-internal ground of the whole or nothing is. The question has a determinate answer, even if we cannot know it.

7.2 The verdicts defined

Verdict	Definition	Plain words
F — Filled	π_1 - π_3 granted, and exactly one x satisfies R(x)	One non-internal ground of both sides
S — Split	π_1 - π_3 granted, no unique occupant, but something satisfies R ⁻ (x)	Something outside grounds one side, or the sides have different grounds, or several things share the job

Verdict	Definition	Plain words
E — Empty	$\pi_1\text{--}\pi_3$ granted, and nothing satisfies $R^-(x)$	Nothing outside grounds either side
R — Refused	At least one of $\pi_1\text{--}\pi_3$ denied	The question misfires

Split covers five distinct situations. It helps to name them.

Split subtype	Situation	Example
SE	Existence grounded from outside; lawfulness brute or internal	A creator of stuff whose regularities are a further brute fact
SL	Lawfulness grounded from outside; existence brute or eternal	Aristotle's unmoved mover; Nyāya's Īśvara over eternal atoms
S ₂	Different outside grounds for each side	A world-maker plus independent Forms
S _{pl}	Several items that <i>jointly</i> ground, and none alone	Plural co-eternal principles
S ^c	Several items, each a full ground: the role is overcrowded	Two independent creators, each sufficient

7.3 The decision procedure

The verdicts come from three yes-or-no questions asked in order. Figure 7.1 draws them.

1. Does the view grant π_1 , π_2 and π_3 ? If **no**, the verdict is **R**.
2. Does the view posit anything non-internal that grounds* E(P) or L(P)? If **no**, the verdict is **E**.
3. Is there exactly one such thing, and does it ground both? If **yes**, the verdict is **F**. If **no**, the verdict is **S**, with a subtype from the table above.

Figure 7.1 · The decision procedure

Proof that the four verdicts are exhaustive and exclusive. Each question has exactly two answers, and each answer leads either to a verdict or to the next question. Any view that gives a determinate answer to all three questions therefore reaches exactly one verdict. ■

A view that is *indeterminate* on a question — for example, a tradition with two live readings — gets a disjunctive verdict, such as “F or S,” with the readings named. This is not a fifth verdict. It says that *the view* has not decided. The infographic’s own maps, which say “some deist pictures” and “some Taoist readings,” already do this informally.

Figure 7.2 redraws the four verdicts in the infographic’s visual language.

Figure 7.2 · The four verdicts as arches, in the infographic’s visual language

7.4 “I don’t know” is a profile, not a verdict

The infographic rightly says “Not knowing is unfinished inquiry; it is not Empty and not a fifth verdict.” The framework can make that precise. A *person* can hold a **verdict profile**: degrees of confidence across the four verdicts that add up to 100%. For example:

- a convinced classical theist: F 95%, S 2%, E 2%, R 1%;
- an agnostic: F 35%, S 10%, E 40%, R 15%.

A *view* gets a verdict. A *person* gets a profile. The “agnostic” is then not a fifth kind of view but a spread-out profile, and disagreements can be located exactly: two people may share the same verdict and disagree only about the occupant theory.

7.5 Tags: saying what kind of verdict

A verdict can carry optional tags written as subscripts or in brackets. The tags carry information that the arch diagram leaves out.

- **Species of ground** (Chapter 9):
 - p = productive (creation, emanation);
 - c = constitutive (the order is an aspect or expression of the ground);
 - n = nomic (the ground legislates the laws);
 - a = axiological (a value requires the order).
- **Occupant kind**: personal or impersonal; concrete or abstract.
- **Mode**: one-time (the ground acts once) or sustaining (it holds the order in being continually).

For example, classical Christian theism is F[p, n; personal; sustaining]. Plotinus is F[p (emanative); impersonal; sustaining]. Aristotle is SL[a (final cause); impersonal].

Chapter 8 — Where God* sits in the literature, and what is new

8.1 The nearest neighbours

None of the ingredients of God* is new. What matters is the combination and the rules for using it. The neighbours fall into four families. Figure 8.1 maps them.

1. “God” as a title or role.

- **Nelson Pike (1969)** read “God” as a title term. His title carried the full perfect-being job description: omnipotent, omniscient, perfectly good. God* keeps the title form and replaces the job description with a thin cosmological one.
- **David Lewis (1970)** supplied the general method (§2.2).

2. “Ground” and “foundation.”

- **Paul Tillich** called God the “ground of being” and denied that God is a being (→ A.14). God* is compatible with Tillich’s view, as F[c, impersonal], but does not require it.
- **Recent analytic debates** frame theism versus naturalism as a dispute about the *foundation* of reality. Examples are Oppy (2013; 2018), the Rasmussen and Leon dialogue *Is God the Best Explanation of Things?* (2019), and Oppy and Pearce’s *Is There a God?* (2022). God* is a sharpened version of their shared question.
- **Richard Swinburne’s *The Existence of God* (2004)** treats the cosmological argument, from the existence of a complex physical universe, separately from the teleological argument, from its orderliness and laws. That is the clearest precedent for G-existence and G-law as two explananda. Swinburne treats them as two pieces of evidence for one hypothesis. God* treats them as two *sides of a role*, and so can say that they might have different grounds.

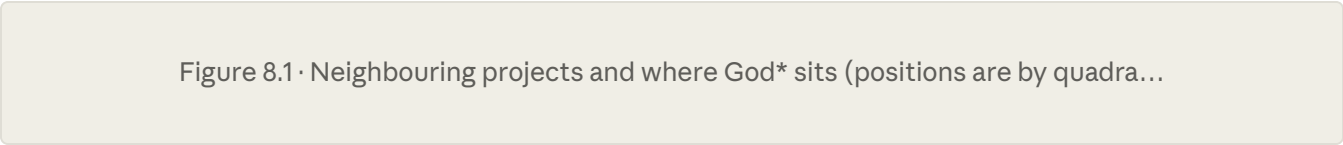
3. Comparative categories across traditions.

- **The Comparative Religious Ideas Project** at Boston University (Neville, ed., 2001) used *vague categories* such as “ultimate realities.” These were deliberately general enough that each tradition could specify them in its own way, and the comparison then examined the specifications. God* is a cousin, but more precise and more falsifiable: it can return Empty or Refused, and its verdicts can be argued with.

- **Wesley Wildman (2010; 2017)** classified “ultimacy models” into families: agential-being theism, ground-of-being theism, and process theism. His families sort views by the *nature* of the ultimate. God* sorts them by whether and how the *role* is filled. The two classifications are orthogonal and complementary.
- **John Hick (1989)** proposed “the Real” as a single reality to which all the great traditions respond. That is a substantive thesis, religious pluralism. God* is only a question, and it is neutral on pluralism.
- **J. L. Schellenberg (2005)** defined **ultimism** as the claim that there is a reality that is ultimate metaphysically, axiologically (in value) and soteriologically (in the good it offers). God* isolates a *relative* of his first dimension, limited to the physical order and split into two sides.

4. Global philosophy of religion.

Kevin Schilbrack’s *Philosophy and the Study of Religions: A Manifesto* (2014), the essays in Draper and Schellenberg’s *Renewing Philosophy of Religion* (2017), and Yujin Nagasawa’s work on global philosophy of religion have urged philosophers to stop treating Christian theism as the default. God* is a tool for that programme.



8.2 What is new

Measured against these neighbours, God* contributes six things. Each is rated for how new it is.

Contribution	Assessment
1. A two-sided role (existence, law) with a named Split verdict	Partly new. The two explananda are old; making them two sides of one role with a verdict for their separation is not standard.
2. Refused separated from Empty by stated presuppositions (Chapter 11)	New as an explicit test in this literature
3. Verdicts score <i>claims</i> , not traditions; “maps” are defeasible tags	New as a method rule. Similar in spirit to comparative religion’s caution.
4. An operational test for “internal” (§12.3)	Partly new. It sharpens Albert’s point into a rule.

Contribution	Assessment
5. Role and occupant strictly separated, with personhood and value excluded from the role	Not new (Lewis, Pike), but rarely enforced this strictly across traditions
6. Verdict profiles, subtypes and tags (§7.4–7.5)	Added in this thesis

Key idea. God* is best presented not as a new theory of God but as a **new instrument**: a shared, precise question with a closed set of answers, which lets rival theories be compared on the same job. That is a publishable kind of contribution in philosophy of religion. Appendix D says how.

8.3 Clashes of terminology to fix

Three terms in the infographic collide with established usage. A referee would notice all three.

1. **“Internal.”** Rudolf Carnap (1950) distinguished *internal* questions, asked inside a linguistic framework, from *external* questions about whether to adopt the framework. He counted external questions as pseudo-questions unless they are practical. The infographic’s “internal” means “presupposing the physical order,” which is related but different. A Carnapian would say the God* question is an external one, and would give the verdict **Refused**. God* v2 therefore uses **intra-physical** in formal statements and keeps “internal” as shorthand.
2. **“Ground.”** In current metaphysics, “grounding” names a *specific* non-causal relation. God* needs something broader. v2 writes **ground*** and defines it (Chapter 9).
3. **“Empty.”** In philosophy of language an “empty name” is one with no referent, which is compatible with the infographic’s use. It should not be confused with Buddhist *śūnyatā*, “emptiness,” which belongs under **Refused**. v2 adds a footnote saying so.

Part III — Stress tests and their resolutions

This part takes the four problems raised in the original evaluation, plus the others a referee would raise, and resolves each. Each resolution ends in a boxed **Ruling**: the change it makes to the framework.

Chapter 9 — What “ground” must mean

9.1 The problem

The infographic’s footnote defines ground as “that in virtue of which the whole order is there and lawful, rather than there being no such order; not a cause inside the order.” The phrase “in virtue of” is the vocabulary of **metaphysical grounding** (§2.4). The main examples of occupants, however, are creators, and creation is usually treated as a kind of **causation** or **production**. Each reading, taken alone, biases the framework.

If “ground” means **strict metaphysical grounding**, three things go wrong.

- *Necessitation*. A full ground guarantees what it grounds. But classical theism holds that God created *freely*: God could have refrained. If God’s nature or will *necessitated* the world, the world would be necessary. That is the modal collapse of §2.6. So on the strict reading a free creator is not a ground, and classical theism could not fill its own role.
- *Constitution*. Paradigm grounds are *constituents* or *more basic aspects* of what they ground, as members are to a set. A creator is not a constituent of the world. The strict reading would therefore favour Spinoza and pantheism.
- *Relata*. Many grounding theorists relate facts, not things. God* speaks of a “whatever” that grounds. A little adjustment handles this: take the fact that the occupant exists, or acts.

If “ground” means **causation**, the opposite biases appear.

- Causation is usually taken to link events in time. That conflicts with the infographic’s own constraint that ground “is not an earlier event.”
- Emanation (Plotinus), constitution (Spinoza, Advaita) and axiological requirement (axiarchism) are not causes in the ordinary sense, so their occupants would be excluded.

9.2 The resolution: ground*, a generic relation with four species

Jessica Wilson (2014) argued that “Grounding” with a capital G is a generic umbrella over many specific “small-g” dependence relations, and that real explanatory work is done by the specific relations. This thesis takes that point constructively.

Definition. *x grounds** a fact *f* if and only if the following five conditions hold.

1. **Explanation.** *f* obtains at least partly *because of* *x*. *x* figures in a correct, non-trivial explanation of why *f* obtains rather than not.

- 2. **Asymmetry.** f does not in the same sense explain x.
- 3. **Species.** The explanation proceeds by one or more of the four species below.
- 4. **No necessitation required.** x need not guarantee f. A free or non-necessitating source counts.
- 5. **No temporal priority required.** x need not come earlier in time than f, and need not be in time at all.

The four species are these.

Species	Tag	How x explains the order	Examples
Productive	p	x brings about, emanates or conserves the order	Abrahamic creation and conservation; Plotinus’s emanation; Nyāya’s Īśvara arranging atoms
Constitutive	c	the order is an aspect, mode, body or appearance of x	Spinoza’s substance; Advaita’s Brahman; Rāmānuja’s world as the body of Brahman; Tillich’s being-itself
Nomic	n	x fixes or legislates the laws	Divine decree (Descartes, Boyle); Plato’s Forms as patterns; Zhu Xi’s <i>li</i>
Axiological	a	x’s value requires or attracts the order	Leslie’s axiarchism; Plato’s Good; Aristotle’s mover as final cause

Figure 9.1 draws the tree.

Figure 9.1 · Four species of ground*

Why this is not cheating. A generic relation might seem to let anything count. The limits come from the other clauses: asymmetry, non-internality, and the requirement that the explanation be *correct and non-trivial by the view’s own lights*. God* does not decide which explanations are true. It records which ones a view claims. Mere correlation is not enough. Neither is being “the same age as the world.” The view must hold that f obtains *because of x*.

One technical warning. Counterfactual tests, of the form “if x had not existed, f would not have obtained,” fail for necessary beings. A counterfactual whose antecedent is impossible is trivially true on the standard semantics. Ground* is therefore defined through *explanation*, which is hyperintensional, not through counterfactuals.

9.3 The terminal clause: which ground is the occupant?

In many systems, several non-internal items stand in a chain. Plotinus has the One, then Intellect, then Soul, then the world. Some Christian theologians hold that God’s ideas ground the laws and God grounds God’s ideas. By transitivity, every link could count as a ground* of P , and the role would look overcrowded (S^c).

Ruling 9.3. The occupant is the **terminal** non-internal ground*: the one that grounds* both sides and is not itself grounded* by a further non-internal item. Intermediate links are *channels*, recorded in the tags but not counted as rival occupants. If the chain has no terminus — an infinite regress of non-internal grounds — there is no unique occupant, and the verdict is Split, written S^∞ .

This also settles the Euthyphro-style case, where God is good because of the Good. If the view holds that the Good grounds God’s nature and God grounds the world, the Good is terminal, and the occupant is the Good with God as channel. If instead God and God’s goodness are identical, as divine simplicity holds, there is one occupant.

Ladder — ground*

Rung 1. There are different ways one thing can be the reason for another. A baker makes a cake. A cake is made *of* flour and eggs. A recipe says how the cake should be. And a birthday is the reason there is a cake at all. God* allows any of these kinds of reason, as long as it comes from outside the world.

Rung 2. “Ground” here means “the thing that explains why the world is there and orderly.” That could be:

- a maker (productive);
- something the world is an expression of (constitutive);
- a rule-giver (nomic);
- a value or purpose (axiological).

It does not have to be earlier in time, and it does not have to force the world to exist.

Rung 3. Ground* is a determinable, a general kind, whose determinates are productive, constitutive, nomic and axiological dependence. It is asymmetric and explanatory. It does not require necessitation or temporal priority. It is broader than Fine-style grounding, which requires necessitation, and narrower than “any explanation,” because it needs a non-internal explainer and a named species.

Rung 4. This is a pluralist, Wilson-style reading of “Big-G.” Open problems:

- whether constitutive ground* can be non-internal at all. Chapter 10 answers yes, where the grounding item has aspects not exhausted by P.
- whether axiological explanation is a genuine species of explanation. Leslie and Rescher say yes; critics call it a category mistake.
- how ground* behaves under transitivity across species. Ruling 9.3 finesses this with the terminal clause.

Ruling 9. Replace “ground” with **ground***, defined as above. Every verdict of F or S carries a species tag.

Chapter 10 — Hard cases

Each case follows the same pattern: the case, why it is hard, the resolution, and the verdict.

10.1 The simulation hypothesis

Case. Nick Bostrom (2003) argued that we may live in a computer simulation run by a more advanced civilization. The simulator is outside *our* physical order and appears to be responsible for both its existence and its laws. Does the simulator fill God*?

Why it is hard. Under the infographic's wording, the simulator looks non-internal to our order and grounds both sides. It would be absurd if a programmer in a basement filled the role whose historical candidates include the One and the Necessary Existent.

Resolution. The phrase “the physical order *as a whole*” must mean the **maximal** physical order, P*. That is everything linked to our paradigm physical objects by spatiotemporal relations or by law-governed interaction (Chapter 12). A physical simulator interacts, under laws, with the physical hardware on which our world runs. That hardware is part of the base world, and our world supervenes on it. So the simulator belongs to P*, and is internal. It is a *local* ground of our universe, not a ground of P*.

Verdict. The simulation hypothesis is **verdict-neutral**. It moves the question down one floor. If the base world is physical, the question about P* remains exactly as open as before. Only if the base level is a non-physical mind would the simulator be a candidate occupant. That is Berkeley's view with extra steps (§10.7). David Chalmers's *Reality+* (2022) makes a related point from the opposite direction: a simulated world is still a real world.

10.2 Spinoza

Case. Spinoza says there is one substance, *Deus sive Natura*, “God or Nature.” It has infinitely many attributes, of which we know two: extension (the physical) and thought. Every particular thing is a *mode* of it. God is “the immanent, not the transitive, cause of all things” (*Ethics* I, prop. 18).

Why it is hard. Is Spinoza's God the physical order itself, which would make it internal and the verdict Empty? Or is it more than the physical order, which would make it non-internal and the verdict Filled?

Resolution. Spinoza distinguishes *natura naturans*, nature as active and generating (substance and its attributes), from *natura naturata*, nature as generated (the modes). The physical order is substance *under the attribute of extension*, in the form of its modes. Substance itself has infinitely many further attributes and is not a part of P. Its nature is not stated in P's vocabulary alone. By the test in §12.3 it is **non-internal**, even though it is not “elsewhere.” Non-internal does not mean spatially outside. It means not part of, and not exhausted by, the physical order.

Verdict.

- On the standard reading of Spinoza, **F[c, n; impersonal; sustaining]**. Laws follow from the attributes’ natures, so the species include nomic.
- On a *naturalist* reading, which keeps only extension and treats “God” as an honorific for physical nature, **E (identity)**.

The framework shows exactly where the two readings of Spinoza part company.

10.3 Priority monism

Case. Jonathan Schaffer (2010) argues that the cosmos as a whole is the one fundamental thing, and that its parts are grounded in it, rather than the whole in its parts.

Resolution. Priority monism is about grounding *within* P, the whole over its parts. The whole is P itself and cannot be P’s non-internal ground.

Verdict. E (whole-first). A useful lesson follows: *fundamental* is not the same as *non-internal ground*. The infographic makes the same point: “Being fundamental among physical processes does not fill the role.”

10.4 A necessary cosmos

Case. Some naturalists, notably Graham Oppy, suggest that the initial state of the natural world, or the natural world as a whole, might be *necessary*. It then needs no outside explanation, and the PSR is satisfied without God.

Why it is hard. Necessity is a kind of explanation: “it had to be.” Is a necessary cosmos therefore an “outside account”?

Resolution. Necessity is a *modal status* of P, not an item outside P. A necessary P is explained, if at all, *by its own nature*, which is internal. The infographic’s “necessity given the laws does not fill the role” extends naturally: **no grade of necessity of P fills the role**. What necessity does is make the role *idle*. That is one way of saying nothing fills it.

Ruling 10.4. Empty has four named subtypes. Recording them shows how a naturalist reaches Empty, which the original infographic left undivided.

Subtype	Claim	Example
E-brute	P exists and is lawful for no reason	Russell’s “just there”; Sean Carroll’s “the universe just is”

Subtype	Claim	Example
E-necessary	P, or its initial state, could not have failed to exist	Oppy; some readings of Tegmark
E-regress	an infinite internal chain, each state explained by an earlier one	Past-eternal cyclic models taken as complete explanation
E-identity	the proposed ground just <i>is</i> P or an aspect of it	Stoicism; naturalist Spinoza; priority monism; Tegmark

10.5 Advaita Vedānta

Case. For Śaṅkara, Brahman alone is ultimately real (*pāramārthika*). The world is real at the conventional or transactional level (*vyavahārika*) but is an appearance (*vivarta*) superimposed on Brahman through *māyā*. The *Brahma Sūtras* define Brahman as “that from which the origin, etc., of this [world] proceed” (1.1.2). Filled, or Refused?

Why it is hard. At the ultimate level, there is no world to ground, which sounds like denying π_1 . At the conventional level, Brahman is plainly the world’s source.

Resolution. The physical order is located at the conventional level, and Advaita *answers* the role question there: Brahman is both the material and the efficient cause of the world. The claim that the world is ultimately an appearance is an occupant-theory claim about the *status of what is grounded*. It is not a refusal of the question.

Ruling 10.5. Score a multi-level tradition *at the level where it locates the physical order*, if it answers the role question at that level. Record other levels in a note.

Verdict. Advaita: F[c, p; impersonal (nirguṇa) or personal as Īśvara (saguṇa); sustaining], with the note “the grounded is ultimately non-different from the ground.” Rāmānuja: F[c; personal]. Madhva: F[p; personal].

Contrast. Most Madhyamaka readings are still **R**. The reason is not that emptiness is ultimate. It is that, *even conventionally*, the Buddha set aside questions about whether the world as a whole is eternal or has an origin as *avyākṛta*, “undetermined,” and Nāgārjuna argued that the concept of a self-standing ground is incoherent. Buddhist thinkers such as Dharmakīrti argued in detail against a creator Īśvara and *did* engage the question. Those schools are **E**, matching the infographic’s map.

10.6 Axiarchism: can a value fill the role?

Case. John Leslie (1979) and Nicholas Rescher (2010) propose that the world exists *because it is good that it should*. The ethical requiredness of a good universe is itself creatively effective. There is no person, no mind and no substance, only a value doing explanatory work. Plato's Form of the Good is the ancient ancestor.

Why it is hard. Can an abstract value “ground” anything? It seems to be the wrong category of thing.

Resolution. The phrase “whatever, if anything” was written for this case. Under Ruling 9, *axiological* ground* is one of the species. Whether axiological explanation really works is a question about the occupant theory, which critics may reject. It does not bar the role.

Verdict. F[a; abstract; impersonal]. This is the clearest example of a Filled verdict with no god in any ordinary sense.

10.7 Idealisms

Case. Idealists hold that the physical world depends on mind. There are several varieties.

Variety	Claim	Verdict
Berkeley	Physical objects are ideas sustained in God's mind; laws are God's regular way of producing them	F[p, n; personal; sustaining]
Analytic idealism (Kastrup)	The physical world is how a universal mind's activity appears from outside; mind is prior and not physical	F[c; impersonal mind]
Cosmopsychism (Goff)	The cosmos is itself conscious, and consciousness is the intrinsic nature of physical structure	E-identity, since the mind <i>is</i> P's nature; F[c] only if the cosmic mind is posited as prior to and independent of physical structure
Finite-minds idealism without God	The physical is a construction from many finite minds' experiences	S _{pl} , if the minds are non-internal and jointly ground P; E if the minds are themselves part of the order

Lesson. The same word, “mind,” can fill, fail to fill, or share the role, depending on whether the mind is *outside* the order (prior, not specified in physical terms) or *is* the order’s inner nature.

10.8 The mathematical universe

Case. Max Tegmark (2008) holds that the physical universe *is* a mathematical structure, and that all mathematical structures exist physically.

Resolution. Tegmark proposes an *identity*, not a dependence. By the terminal-clause logic of §9.3, nothing can be the non-internal ground of what it is identical to.

Verdict. **E-identity** (with E-necessary as a secondary tag, since mathematical existence is necessary). A different view, on which mathematical structures exist independently and *make it the case* that one of them is physically real, would be **F[c or n; abstract]**. The ancient Pythagoreans, who said that “all is number,” sit between these two.

10.9 Deism and the “starter”

Case. The infographic places “a starter after which law needs no further ground” under Split.

Why it is a problem. The typical eighteenth-century deist God creates matter *and* laws, then leaves them to run. That is one non-internal source for both sides. It makes no difference that the ground acts once rather than continually. The infographic’s constraint, “ground is not an earlier event in time,” is about the *occupant*, not the *mode* of its action. The occupant is the non-internal agent, not the event of starting.

Ruling 10.9. Grounding once and grounding continually are *modes*, recorded as tags. They are not different verdicts.

Verdict.

- Classical deism: **F[p, n; personal; one-time]**.
- A starter that produces the world’s stuff but whose laws are self-standing and not produced: **SE**.
- A starter who only orders pre-existing stuff: **SL**.

The infographic’s map should therefore say “some deist pictures, where God starts the stuff but not the laws.” Most deists land in Filled.

10.10 Process theism

Case. In Whitehead's *Process and Reality* (1929), God is not before all creation but *with* all creation. God's "primordial nature" envisages all the pure possibilities ("eternal objects") and offers each moment of experience its initial aim. That supplies order and novelty. But "creativity," the ultimate fact that there are ever new occasions, is not God's product. Later process theologians describe God's creating as "out of chaos," not out of nothing (Cobb and Griffin 1976).

Verdict. **SL[n, a; personal; sustaining]**. God grounds* order. The existence of a creative process is ultimate and independent. Process theism is therefore a modern theistic Split, which shows that Split is not a merely antique category.

10.11 Plato, Aristotle and Nyāya: Split confirmed

The framework's tools confirm the placements of Chapter 3.

- **Plato (*Timaeus* reading):** **S₂[p, n]**. There are two non-internal grounds of the law-side, the Demiurge and the Forms, and none of the receptacle's existence.
- **Plato (*Republic* reading, with the Good as source of all being):** **F[a, n]**.
- **Aristotle:** **SL[a; impersonal]**.
- **Nyāya-Vaiśeṣika:** **SL[p; personal]**, since the atoms are eternal and uncreated.

10.12 Gods inside the cosmos

The Stoics' fiery logos, the Epicurean gods, and Hesiod's gods, born from Chaos and Earth, are all **E-identity** or plain **E** with the tag "immanent deities." On a common reading of Latter-day Saint theology, God organizes co-eternal elements (Doctrine and Covenants 93:33) under co-eternal laws. That yields **SL** or **E**, depending on whether the organizing is itself a non-internal ground* of order. As always, the verdict scores the stated claim and makes no judgment of the tradition.

10.13 Dualist systems

Manichaeism posits two co-eternal principles, Light and Darkness, whose mixture produces the physical world. That is **S_p**: two principles jointly, and neither alone. Some **Gnostic** systems have a lower craftsman, the Demiurge, who shapes matter, and a higher, unknowable Monad from which everything, including the Demiurge, ultimately flows. By the terminal clause, that is **F** with the Monad as occupant, unless matter is held to be independent of the Monad, which gives **S₂**.

10.14 Evolving laws

C. S. Peirce suggested that laws of nature evolved as “habits” of the universe. Lee Smolin’s *cosmological natural selection* has laws that vary through black-hole “reproduction.” Both explain lawfulness by a process *within* P* — a meta-law or a habit-forming tendency — so both are **E-regress** or **E-brute** at the level of the meta-law.

10.15 Zhu Xi, and readings that genuinely differ

Zhu Xi’s Supreme Ultimate grounds the li of all things. Whether it also grounds the existence of qi is disputed among interpreters.

Verdict. F or SL, with the dispute named. This is the framework working as intended. It does not force a decision that the texts do not make. It shows exactly what question the interpreters are disagreeing about.

Chapter 11 — Refused versus Empty

11.1 The worry

Empty and Refused both say, in effect, “no God*.” The infographic separates them: “Empty says nothing fills the role. Refused says the role misfires. Score the claim, not the tone.” But without a test, the line blurs. Is Hume Empty or Refused? Is Russell? Is Kant? A critic could say the category exists only to be generous to Buddhists and Daoists. The fix is to name exactly what Refused rejects.

11.2 The presupposition test

The role question has three presuppositions (§7.1). A view is **Refused** if and only if it denies at least one of them. Each denial has a distinct historical family.

Presupposition	Denial sounds like	Who denies it
π_1 , totality: there is a physical order that can be spoken of as a whole	“There is no ‘it’ to ground”	Kant: the world-whole is not an object of possible experience. Nāgārjuna: no self-standing whole.

Presupposition	Denial sounds like	Who denies it
π_2 , explicability: dependence can apply to the whole	“Grounding doesn’t apply to the whole”	Russell: cause is not applicable to the total. Carnap: an “external” pseudo-question. The Buddha’s undetermined questions.
π_3 , determinacy: either something grounds it or nothing does	“Neither ‘grounded’ nor ‘ungrounded’ applies”	The Madhyamaka tetralemma, which rejects all four options (is, is not, both, neither). Verificationism. Some readings of the Dao as beyond all distinctions.

Empty accepts all three presuppositions and answers “nothing.” That includes “nothing, and nothing is needed,” the E-necessary and E-brute subtypes.

Key idea. “It needs no ground” is **Empty**. “The question of its ground is ill-formed” is **Refused**. The first answers the question. The second rejects it.

11.3 A worked example: Russell against Copleston, 1948

In the BBC debate, Russell said several different things, which is why readers disagree about his position.

Russell’s claim, paraphrased	Presupposition touched	Scores as
“The universe is just there, and that’s all.”	none; accepts the question and answers it	E-brute
The concept of cause is not applicable to the universe as a total	π_2	R
“The universe” may not be a meaningful object at all, as the mother-of-the-human-race analogy suggests	π_1	R
The word “necessary” applies only to propositions, not to beings	This bears on the occupant theory, not on the role	no verdict

Ruling 11.3. When a thinker makes both a refusing claim and an emptying claim, record the verdict as **R with fallback E**, written $R \rightarrow E$. It means: “I refuse the question; but if the presuppositions are granted, I say nothing fills the role.” Russell is $R \rightarrow E$. Carnap is R, with no stated fallback. Many modern naturalists who say “the question is confused, and anyway the universe just is” are $R \rightarrow E$.

The fallback notation makes it clear that Refused is not a disguised Empty, while being honest that many refusers would settle at Empty if pressed.

11.4 Refusal, unknowing and practical faith

Three positions are often confused with Refused.

1. **Unknowing.** The Nāsadīya hymn’s “perhaps he does not know” is an epistemic humility about the answer. It is a spread-out *verdict profile* (§7.4), not a verdict.
2. **Kant.** Kant refuses the question as a matter of *theoretical* knowledge: speculative reason cannot reach the world-whole. Yet for *practical* purposes he postulates God. Score each claim: theoretical R, practical postulate F[p, n; personal]. Kant is the clearest example of why the rule “score claims, not thinkers” is needed.
3. **Apophatic theology.** Pseudo-Dionysius and Tillich deny that God is a being, or even that “God exists” in the ordinary sense. That can sound like refusal. But they affirm that God is the source of all that is. So they are F, with the occupant-theory tag “beyond being.” Tone is not verdict.

11.5 A phrasebook

If a view says...	Verdict
“X, beyond the physical, makes it exist and makes it orderly.”	F
“X orders it; its existence is eternal or brute.”	SL
“X gives it existence; its order is brute or self-standing.”	SE
“X and Y together are why it exists and is orderly.”	S_{pl} or S_2
“Nothing outside it; it just is.”	E-brute
“Nothing outside it; it had to be.”	E-necessary
“What you call its ground is just the world itself.”	E-identity

If a view says...	Verdict
“There’s no ‘whole’ to ask about.”	R (π_1)
“Grounding and explanation don’t apply to the whole.”	R (π_2)
“None of ‘grounded,’ ‘ungrounded,’ both or neither is right.”	R (π_3)
“We can’t know.”	A verdict profile, not a verdict
A ritual, a practice or a way of life that makes no metaphysical claim	Not scored (§13.7)

Ruling 11. Refused is defined by the denial of π_1 , π_2 or π_3 . Mixed positions are recorded as $R \rightarrow E$, or, rarely, $R \rightarrow F$.

Chapter 12 — The scope of “physical”

12.1 Why the physical order, and not everything?

A critic asks: “Why only the *physical* order? Dualists think minds are non-physical. Platonists think numbers are. Why not ask for the ground of *everything*?” There are two answers.

The totality problem. Nothing is outside *everything*. If the role were “the non-internal ground of all that exists,” the occupant would itself be part of all that exists, and the role would be unfillable by definition. That would be a hidden bias toward Empty. Classical theists do not say God is outside reality. They say God is outside *creation*. A useful role must pick a domain whose outside is a live question.

The shared datum. The physical order is the one domain that every party accepts. Theists, naturalists, idealists and Buddhists all agree that there *is* a physical world, whatever its ultimate status. Minds and numbers are contested, so building them into the domain would make the vocabulary less shared.

This is why the infographic’s choice is right. Section 12.5 adds optional extensions for people who want a wider domain.

12.2 Hempel’s dilemma, and how to avoid it

Carl Hempel (1969) posed a dilemma for anyone who says “physical.”

- If “physical” means “what current physics describes,” the definition is almost certainly wrong, because current physics is incomplete. Quantum gravity is missing, for example.
- If “physical” means “what the ideal, final physics would describe,” the definition is empty, because we do not know what that physics will include. It might include minds.

The solution used here is Daniel Stoljar’s *object-based* strategy, adapted to God*.

Definition. Start with paradigm physical objects: this table, this electron, that star. The **maximal physical order P^*** is everything connected to the paradigms by a chain of links. Each link is either:

- a **spatiotemporal relation** (being at some distance or time from), or
- a **law-governed interaction**: two items that affect each other and are both subject to the same, or continuous, laws.

The definition does not care what physics says the items are. It only tracks connection. A one-way ground* relation, such as creation, is **not** a link, because it is asymmetric and not itself governed by the laws of what it grounds.

What this gains. It answers Hempel: P^* is fixed by connection to the paradigms, not by the content of any theory. It also answers the simulation case: a physical simulator interacts, under laws, with the physical system that runs our world, so it is in P^* . And it gathers the multiverse, past cycles and parallel branches of quantum mechanics into P^* whenever they are linked by physical law.

Figure 12.1 · Nested orders and the maximal physical order P^*

12.3 An operational test for “internal”

Definition. An item x is **internal** (intra-physical) if and only if at least one of the following holds.

- **(a) Membership.** x is in P^* .
- **(b) Presupposition.** The theory positing x cannot state what x is without using P^* ’s vocabulary: spatiotemporal location, physical magnitudes, or P^* ’s laws and dynamics. “A generator that takes an ordered domain as input” fails here.
- **(c) Identity.** x is identical to P^* , or to an aspect of it whose nature is exhausted by P^* ’s structure.

Otherwise x is **non-internal**.

Candidate	(a)	(b)	(c)	Result
The Big Bang singularity or first state	yes			internal
The inflaton field or multiverse generator	yes			internal
A quantum vacuum or the Hartle–Hawking wave function		yes		internal
The physical laws themselves, on a governing view		yes		internal
A simulator in a physical base world	yes			internal
The Stoic logos	yes			internal
Tegmark's mathematical structure			yes	internal
The cosmic subject of cosmopsychism			yes	internal
Spinoza's substance				non-internal
Advaita's Brahman				non-internal
Leslie's ethical requirement				non-internal
The classical creator				non-internal
The Platonic Forms				non-internal

One entry deserves comment. On a *governing* view, the laws are internal because they are laws of P^* . They are stated in its vocabulary. This is why “the laws explain the world” is never a Filled verdict. The laws are part of what G-law asks about.

12.4 Minds and numbers

If minds or numbers are non-physical, does that cause trouble? No. Being non-internal makes something *eligible* to fill the role. It does not make it an occupant. The number 7 is non-internal, but no one thinks it grounds the physical order. A Cartesian soul is non-internal, but it grounds only its own thoughts. The Pythagoreans who said “all is number” did propose an occupant of the abstract kind, which is scored under §10.8. Non-physical items raise no problem unless a view says they ground P^* , and then they are scored like any other candidate.

12.5 Optional wider roles

For research that needs them, God* can be widened.

- **God*^c**: the non-internal ground of the whole *concrete* order, physical *and* mental.
- **God*^A**: the non-internal ground of the concrete order *and* of abstract objects such as numbers and propositions. This role is close to the traditional “God as creator of all things visible and invisible.”

The verdicts often stay the same. Berkeley is F in all three roles. The Stoics are E in all three. Some views change. A theist who holds that numbers exist independently of God is F for God* and God*^c, but S for God*^A. That is exactly the point where “theistic activism,” the view that God creates abstract objects too, is debated. The extensions make such differences visible without complicating the main role.

Ruling 12. “The physical order as a whole” means **P***, the maximal connected physical order. “Internal” is defined by tests (a) to (c). Wider roles are optional modules.

Chapter 13 — Further objections

13.1 Is there really such a thing as “the whole”?

Objection. Hume said grouping the parts into a whole is an “arbitrary act of the mind.” *Mereological nihilists* deny that composite objects exist at all. So “the physical order as a whole” may name nothing.

Reply. The role does not need a single *object* called “the whole.” George Boolos (1984) showed that *plural* logic can speak about “the physical things, taken together” without positing an extra thing, the set or sum, that contains them. G-existence then becomes a plural fact: *there are some physical things*. That fact is not about any one of them, so explaining each one does not explain it. Hume’s twenty particles are each explained, but the fact that there are any particles at all is a further fact. This does not prove the further fact *has* an explanation. Empty is always allowed. It shows only that the question is well-formed.

13.2 Is this the fallacy of composition?

Objection. Russell’s mother argument: every part has a cause, but it does not follow that the whole does.

Reply. Correct, and God* does not infer that it does. God* is a *question*, not an argument. The fallacy would be the inference “each part has a ground, so the whole has one.” The framework makes no such inference and treats E-brute as a first-class answer.

13.3 “Why these laws?” — the specificity problem

Objection. G-law says the domain is lawful. What about *which* laws hold, and *which* initial conditions? A ground that makes there be some law, but not these laws, fits no verdict cleanly.

Ruling 13.3. Both sides include their specific content.

- **G-existence** covers *that* there is a physical order *and* its basic inventory and initial state.
- **G-law** covers *that* the order is lawful *and* which laws hold.

A ground that covers only the generic facts, and leaves the specific ones brute, is recorded with the tag “generic only.” This brings Parfit’s “Why this?” and fine-tuning inside the framework without adding a third side.

13.4 Counting grounds: one or many?

Objection. Christianity’s God is three persons. Vedānta’s Brahman has many qualities. Plotinus has a chain of principles. When is there “one” ground?

Reply. Count grounds by the view’s *own* identity criteria for the explanatory source, together with the terminal clause (Ruling 9.3). Classical Trinitarian theology teaches that the works of the Trinity toward creation are undivided (*opera Trinitatis ad extra indivisa sunt*): one act, from one God. So it is F. A pantheon of creator gods that each made part of the world is S_{pl} . Counting is the view’s business. The framework only records it.

13.5 Is God* just the cosmological argument in disguise?

Objection. The role is shaped like the conclusion of the cosmological argument, so it rigs the game toward theism.

Reply. The role is shaped like the *question* the cosmological argument answers. Naturalists ask the same question and answer it differently. “The universe just is” is a direct answer to it. God* assumes no PSR, allows E-brute, and treats Refused with respect. The opposite complaint also comes up: that God* strips theism of everything theists care about. That shows its neutrality. God* is not a replacement for anyone’s God. It is a shared question they can all be scored against.

13.6 The occupant’s own ground

Objection. What grounds God*? If nothing, why can’t the physical order be the ungrounded thing?

Reply. That is exactly the question that separates F from E-brute and E-necessary, and God* leaves it open. The framework is about P* only. Whether the occupant is necessary, self-explaining or brute belongs to the occupant theory. It is the right place for the old debate between the theist's "God is necessary" and the naturalist's "then why not the world?"

13.7 Religions are not only propositions

Objection. Wittgensteinian philosophers of religion such as D. Z. Phillips argue that religious language often does not make metaphysical claims at all. It expresses a way of life. Scoring traditions as if they were theories distorts them.

Reply. God* scores *claims*. A practice, a ritual or a way of speaking that makes no claim about the ground of the physical order is simply **not scored**. That is not a verdict, not even Refused. Refused is an explicit claim that the question misfires. The infographic's own footnote 5 already says that tags show where a tradition often lands, not a claim about every adherent. God* v2 makes "not scored" an explicit category outside the verdicts.

13.8 Who gets to say what a tradition claims?

Objection. The "maps" risk misreading traditions from outside.

Reply. Four protocol rules (Chapter 14) handle this:

1. Cite primary texts or recognized commentators for every map.
2. Name the reading, as in "on Śaṅkara's reading."
3. Allow disjunctive verdicts where readings differ.
4. Treat a map as defeasible, open to challenge by insiders who cite better textual evidence.

13.9 The name "God*"

Objection. Using the word "God", even starred, may alienate atheists and non-theistic traditions, and may annoy theists who feel their word has been taken.

Reply. The star signals a stipulated, technical term. The word was chosen because the dispute it names is the one people usually carry on with the word "God." In settings where the name is a distraction, the neutral label **G*** or "the ground role" can be used. Nothing depends on the name.

Part IV — Using the framework

Chapter 14 — A scoring protocol and an atlas of views

14.1 The protocol

Scoring a view takes seven steps. Each step points back to the chapter that justifies it.

1. **State the claim.** Quote or closely paraphrase the view's claim about why there is a physical order and why it is lawful. Cite a primary text or a recognized commentator. No claim means **not scored** (§13.7).
2. **Fix the level.** If the view has several levels of reality, score it at the level where it places the physical order (Ruling 10.5).
3. **Test the presuppositions.** Does the view deny π_1 , π_2 or π_3 ? If so, the verdict is **R**. Record any fallback ($R \rightarrow E$ or $R \rightarrow F$) (Chapter 11).
4. **List the candidate grounds.** Apply the internal test (a)–(c) to each (§12.3). Discard internal candidates, noting E-identity where the view equates its “ground” with P^* .
5. **Check ground*.** For each non-internal candidate, ask which side it grounds* (E, L or both) and by which species (p, c, n, a). Apply the terminal clause (Chapter 9).
6. **Read off the verdict.** Use the decision procedure (Figure 7.1). Add the subtype (S_E , S_L , S_2 , S_{PI} , S^c , S^∞ ; E-brute, E-necessary, E-regress, E-identity) and the tags (species; personal or impersonal; one-time or sustaining; “generic only”).
7. **Record the uncertainty.** Name rival readings and give a disjunctive verdict where they differ. Note the evidence an insider would need to overturn the map.

14.2 Two worked scorecards

Thomas Aquinas

Step	Finding
Claim	God is <i>ipsum esse subsistens</i> , subsistent being itself, the first cause of the being of all things (<i>Summa Theologiae</i> I, q. 44, a. 1), and governs all things by the eternal law (I–II, q. 93)
Level	One level
Presuppositions	All granted

Step	Finding
Candidates	God, non-internal. Angels are non-internal but are creatures, so they are channels at most.
Ground*	Productive (creation and conservation) and nomic (the eternal law) for both sides; terminal
Verdict	F[p, n; personal; sustaining]
Uncertainty	Low. A world without a beginning would not change the verdict (<i>On the Eternity of the World</i>).

Nāgārjuna

Step	Finding
Claim	Whatever arises dependently is empty of intrinsic nature (<i>Mūlamadhyamakakārikā</i> 24.18). Nothing arises from itself, from another, from both, or without cause (1.1).
Level	The conventional level, where the physical order sits
Presuppositions	π_1 denied: no self-standing whole. π_3 denied: the four-cornered negation.
Candidates	None considered, because the question is dissolved
Verdict	R (π_1, π_3) , with no fallback: Nāgārjuna would also reject “the whole is ungrounded”
Uncertainty	Moderate. Some readers treat Madhyamaka as a critique of <i>all</i> views, so that even “R” is a view to be relinquished. The framework records R as the stated claim.

14.3 The atlas

The table scores about fifty views using the protocol. Verdicts marked “/” are disjunctive, reflecting genuinely different readings. Every verdict is defeasible. The notation comes from Chapter 7: S_L is a law-side Split, S_2 has two grounds, and S_{pl} has plural co-grounds.

View	Verdict and tags	Basis	See
Atomism (Democritus, Epicurus, Lucretius)	E-brute (gods inside the cosmos)	Eternal atoms and void	§3.1
Heraclitus	E-identity	The logos is the world's own measure	§3.1
Pythagoreans	Between E-identity and F[c or n; abstract]	"All is number"	§10.8
Plato, <i>Timaeus</i> reading	S ₂ [p, n]	Demiurge and Forms; the receptacle is ungrounded	§10.11
Plato, <i>Republic</i> reading	F[a, n]	The Good is the source of being and knowability	§10.11
Aristotle	SL[a; impersonal]	Unmoved mover as final cause; the world is eternal	§3.3
Stoicism	E-identity (immanent deity)	God is corporeal logos within the cosmos	§3.4
Plotinus	F[p (emanative), n; impersonal]	Everything flows from the One	§3.4
Genesis 1, read as the ordering of chaos	SL / F	Scholarly dispute over the creation formula	§3.5
Rabbinic Judaism, Maimonides	F[p, n; personal]	Creation from nothing	§3.5
Classical Christianity (Augustine, Aquinas)	F[p, n; personal; sustaining]	Creation and conservation; eternal law	§14.2
Ash'arī occasionalism (al-Ghazālī)	F[p, n; personal; continual]	God re-creates the world; regularities are God's custom	§4.2
Avicenna	F[p (necessary emanation); intellect]	The Necessary Existent	§3.5
Sikhism	F[p, n; beyond form; sustaining]	<i>Ik Onkar; Mūl Mantar</i>	§3.5

View	Verdict and tags	Basis	See
Vedic <i>Nāsadīya Sūkta</i>	A verdict profile, not a verdict	“Perhaps he does not know”	§11.4
Advaita Vedānta (Śaṅkara)	F[c, p; nirguṇa or saguṇa]	<i>Brahma Sūtra</i> 1.1.2	§10.5
Viśiṣṭādvaita (Rāmānuja)	F[c; personal]	The world is Brahman’s body	§10.5
Dvaita (Madhva)	F[p; personal]	Viṣṇu is creator and distinct	§10.5
Classical Sāṃkhya	E-brute	Eternal prakṛti, no creator	§3.6
Nyāya-Vaiśeṣika (Udāyana)	SL[p; personal]	Īśvara arranges eternal atoms	§10.11
Mīmāṃsā (Kumārila)	E	A beginningless world; arguments against a creator	§3.6
Jainism	E	An uncreated, eternal universe	§3.6
Early Buddhism (undetermined questions)	R (π_2)	The world’s origin is set aside	§10.5
Dharmakīrti and Buddhist anti-creator arguments	E	Engages the question, denies Īśvara	§10.5
Madhyamaka (Nāgārjuna)	R (π_1, π_3)	Emptiness; the tetralemma	§14.2
<i>Daodejing</i> , cosmogonic reading	F[p, c; impersonal]	The Dao gives birth to the One	§3.7
Guo Xiang’s Daoism	E-brute	Things are “self-so,” with no producer	§3.7
Daoism, refusing reading	R (π_3)	The Dao is beyond distinctions	§3.7
Zhu Xi	F / SL	Supreme Ultimate grounds li; qi is disputed	§10.15

View	Verdict and tags	Basis	See
Manichaeism	S_{pl}	Light and Darkness co-eternal	§10.13
Valentinian Gnosticism	F / S_2	Monad as terminal, or independent matter	§10.13
Descartes	$F[p, n; \text{personal}; \text{sustaining}]$	Laws derived from God's immutability	§3.8
Boyle, Newton	$F[p, n; \text{personal}; \text{sustaining}]$	Laws as decrees	§3.8
Spinoza, standard reading	$F[c, n; \text{impersonal}]$	Substance with infinite attributes	§10.2
Spinoza, naturalist reading	E-identity	God as an honorific for extended nature	§10.2
Leibniz	$F[p, n, a; \text{personal}]$	God chooses the best possible world	§3.8
Classical deism	$F[p, n; \text{personal}; \text{one-time}]$	A creator who starts matter and laws	§10.9
Hume	$E / R \rightarrow E$ (disputed)	Philo: matter may contain its own principle of order	§3.8
Kant	Theoretical R; practical F	Antinomies; postulates of practical reason	§11.4
Hegel	$F[c] / E\text{-identity}$	Right-Hegelian versus left-Hegelian readings of Spirit	A.13
Whitehead and process theism	$SL[n, a; \text{personal}]$	God orders; creativity is ultimate	§10.10
Tillich	$F[c; \text{beyond being}]$	Being-itself	§11.4
Logical positivism (Carnap, Ayer)	$R(\pi_2, \pi_3)$	An external or unverifiable question	§8.3
Wittgenstein, <i>Tractatus</i>	$R(\pi_2)$	"The riddle does not exist" (6.5)	A.21

View	Verdict and tags	Basis	See
Russell (1948)	$R \rightarrow E$	Two claims, scored separately	§11.3
Latter-day Saint theology, common reading	SL / E	God organizes co-eternal elements	§10.12
Contemporary analytic theism (Swinburne, Plantinga, Pruss, Craig)	F[p, n; personal]	Creator and sustainer	§3.9
Oppy's naturalism	E-necessary / E-brute	A necessary or brute initial state	§10.4
Sean Carroll's poetic naturalism	E-brute	"The universe just is"	§10.4
Schaffer's priority monism	E-identity (whole-first)	The cosmos is fundamental	§10.3
Leslie and Rescher's axiarchism	F[a; abstract]	Ethical requiredness	§10.6
Berkeley	F[p, n; personal; sustaining]	Ideas in God's mind	§10.7
Kastrup's analytic idealism	F[c; impersonal mind]	The world as the appearance of mind-at-large	§10.7
Goff's cosmopsychism	E-identity / F[c]	Depends on the priority of the cosmic mind	§10.7
Tegmark's mathematical universe	E-identity (E-necessary)	Physical existence is mathematical existence	§10.8
Peirce, Smolin: evolving laws	E-regress / E-brute	Laws explained by an internal process	§10.14
Bostrom's simulation hypothesis	Verdict-neutral	Relocates the question inside P*	§10.1

Figure 14.1 places the atlas on a single map.

Figure 14.1 · The atlas on the verdict map; disjunctive views appear under each...

What the atlas shows.

1. **Every verdict is well populated.** Split in particular is not an exotic curiosity. It includes Aristotle, Nyāya, Plato on one reading, and modern process theism.
2. **The verdicts cut across the categories “religious” and “secular.”** Stoicism and Latter-day Saint theology can land in E or S. Axiarchism and Tillich land in F.
3. **Disjunctive verdicts cluster** where traditions have long-running internal debates: Plato, Zhu Xi, Spinoza, Hegel, Daoism. God* locates those debates precisely.

Chapter 15 — What God* cannot do

A framework is only trustworthy if its limits are stated.

1. **It does not decide which verdict is true.** God* supplies a question and a set of answers. Arguments for any verdict, such as the cosmological arguments for F or the brute-fact defences of E, must be judged on their own merits.
2. **It does not capture what makes a religion a religion.** Worship, ethics, community, practice and salvation all lie outside the role by design. A tradition's verdict is a small fact about it.
3. **The physical/non-physical distinction may itself be culturally specific.** Some Indigenous and animist ontologies do not divide the world into “physical” and “spiritual” at all. For such a view, π_1 may fail as a matter of framing, or the view may be *not scored*. Scoring it anyway would repeat the outsider error the framework was built to avoid.
4. **Maps depend on interpretation.** The protocol reduces this but cannot remove it.
5. **Precision costs simplicity.** v2 is more exact than v1 and harder to put on one page. The best form is two tiers: a simple public card, much like the original infographic, with the technical rulings in an appendix.
6. **It is all-or-nothing about grounding.** A view on which a ground only *partly* or *probabilistically* explains the order has no graded verdict. A natural extension would allow degrees of dependence.

Chapter 16 — Conclusion: God* version 2

16.1 Summary of the rulings

Ruling	Problem	Change to the framework
9	“Ground” was ambiguous between grounding and causation, biasing the verdicts	Ground* : generic explanatory dependence, with no necessitation or temporal priority required; species p, c, n, a
9.3	Chains of non-internal grounds looked overcrowded	The occupant is the terminal ground; others are channels. An endless chain is S^∞ .
10.4	Naturalist routes to Empty were lumped together	Empty subtypes: brute, necessary, regress, identity
10.5	Multi-level traditions resisted scoring	Score at the level where the physical order sits
10.9	Deism was mapped as Split	Once versus continually is a mode tag, not a verdict. Classical deism is F.
11	The line between Refused and Empty blurred	Refused means denying π_1 , π_2 or π_3 . Mixed views are $R \rightarrow E$.
12	“Physical” was vulnerable to Hempel’s dilemma and to simulations	P* : the maximal order connected to the paradigms. Tests (a)–(c) define “internal.”
13.3	“Why these laws?” had no home	Both sides include their specific content; tag “generic only”
13.7	Non-propositional religion was misclassified	“Not scored” is a non-verdict
§7.4	“I don’t know” needed a precise status	Verdict profiles for persons

16.2 God* v2: the revised statement

God* is the title of a role, not the name of a being. The role is filled by whatever, if anything, is the **terminal non-internal ground*** of the **maximal physical order P*** taken as a whole. P* has two sides:

- **G-existence:** that there is such an order at all, with its basic inventory and initial state.
- **G-law:** that it is lawful, and has the laws it has.

Ground* is asymmetric explanatory dependence. It may be productive, constitutive, nomic or axiological, and it need be neither earlier in time nor necessitating.

Something is **non-internal** if it is not part of P*, is not stated in P*'s vocabulary, and is not identical to P* or to an aspect of P*.

A view gives one of four verdicts:

- **Filled:** one terminal ground* of both sides.
- **Split:** something non-internal grounds at least one side, but no unique thing grounds both.
- **Empty:** nothing non-internal grounds either side.
- **Refused:** the view denies that the order can be taken as a whole, that dependence applies to it, or that the question has a determinate answer.

Personal or impersonal, once or continually, necessary or contingent: these belong to the **occupant theory**, not the role. Not knowing is a profile over verdicts, not a verdict. A practice that makes no claim is not scored.

16.3 Recommended changes to the infographic

1. **Wording.** Replace “ground” with “ground*” and add a footnote defining it. Write “the physical order as a whole (P*)” and footnote P*.
2. **Split panel map.** Change “some deist pictures” to “Aristotle · Nyāya · process theism · Plato’s *Timaeus*.” Move classical deism to Filled.
3. **Empty panel.** Add the four subtypes as small chips: brute, necessary, regress, identity. Consider drawing the arch as a corbelled arch, which stands with no keystone.
4. **Refused panel.** Add the three presuppositions as three short lines, and the $R \rightarrow E$ notation.

5. **“Internal vs non-internal” box.** Add the simulation example and the rule that the simulator is inside P*.
6. **Constraints box.** Add “Whether the ground acts once or continually is not part of the role” and “A practice that makes no claim is not scored.”
7. **Filled panel map.** Add “Plotinus · Spinoza (standard reading) · axiarchism · Tillich” to show that Filled can be impersonal and even non-theistic.

16.4 Closing

The question “why is there a lawful world at all?” has been asked in Greek, Sanskrit, Pali, Chinese, Hebrew, Arabic and Latin, and answered in many ways. The word “God” has too often stood in the way of seeing the answers side by side. God* does not answer the question. It holds the question still, so that a Thomist, a Madhyamaka Buddhist, a Stoic, a Spinozist and a physicist can each say exactly what they think fills the keystone slot — or why nothing does, or why there is no slot — and be understood by the others. That is a modest goal, but a real one.

Appendix A — Primers

Each primer gives the background the main text assumes, in the order the main text first needs it. Dates are approximate for ancient figures.

A.1 The Presocratics (sixth to fifth centuries BCE)

The first Greek philosophers lived around the Aegean before Socrates, hence “Presocratic.” Their works survive only as fragments quoted by later writers.

- **Thales of Miletus** (c. 585 BCE) said everything is, or comes from, water. What mattered was the idea that one natural principle underlies everything.
- **Anaximander** replaced water with the *apeiron*, the boundless or indefinite.
- **Heraclitus of Ephesus** (c. 500 BCE) taught that everything flows, like a river you cannot step into twice, but that the flux obeys a *logos*, a rational proportion.
- **Parmenides of Elea** (c. 475 BCE) argued that what-is cannot come from what-is-not. That principle sets the terms of every later debate about creation.
- **The atomists** Leucippus and Democritus (fifth century BCE) held that reality is indivisible bodies moving in void. **Epicurus** (341–270 BCE) and the Roman poet **Lucretius** (*On the Nature of Things*, c. 55 BCE) turned atomism into an ethical

philosophy that frees people from fear of the gods, who exist but take no part in the world.

A.2 Agrippa and ancient skepticism

Agrippa (first century CE?) is known through the later skeptic Sextus Empiricus. He listed “modes” for suspending judgment. Three of them — infinite regress, circularity and arbitrary assumption — form the trilemma of §1.3. In the twentieth century Hans Albert renamed it the “Münchhausen trilemma,” after the baron who claimed to have pulled himself out of a swamp by his own hair.

A.3 Plato: the craftsman, the Forms and the Good

Plato (c. 428–348 BCE) was Socrates’ student and Aristotle’s teacher. He wrote dialogues and founded the Academy in Athens.

The Forms. Plato held that beyond the changing things we see there are eternal, unchanging Forms: Beauty itself, Justice itself, Equality itself. Particular things are what they are by “participating” in the Forms. A beautiful face is beautiful by sharing in Beauty. The Forms are known by reason, not by the senses. In the *Republic*’s allegory of the cave, ordinary life is watching shadows on a wall, and philosophy is turning round toward the Forms and finally the sun.

The Good. In the *Republic* (Book VI, 508–509), the Form of the Good is compared to the sun. The sun makes things visible and makes them grow. The Good makes the Forms knowable and gives them their being, while itself being “beyond being in dignity and power.” This passage inspired Neoplatonism (A.5) and much later theology.

The *Timaeus* and the Demiurge. In the *Timaeus* (c. 360 BCE), a character called Timaeus tells a “likely story” of how the cosmos came to be. A good divine craftsman, *dēmiourgos*, meaning “public worker” or “artisan,” wanted everything to be as good as possible. He looked to an eternal model, the Forms, and brought order to a pre-existing, chaotic “receptacle,” which Plato also calls a “nurse of becoming.” He shaped it using geometry: the elements are built from triangles. The craftsman is not omnipotent. He persuades *necessity* rather than overpowering it.

Why it matters for God*. Plato gives *three* candidates: the craftsman (a productive, law-giving orderer), the Forms (a nomic pattern) and the Good (an axiological source). None of them creates the receptacle. That is why Plato is either Split or Filled depending on how one relates the three. Scholars since antiquity have debated whether the *Timaeus* describes a creation in time or is a teaching device for eternal dependence.

A.4 Aristotle: four causes, an eternal world, an unmoved mover

Aristotle (384–322 BCE) studied at Plato's Academy for twenty years, tutored Alexander the Great, and founded the Lyceum. He wrote on nearly every subject.

- **“Metaphysics.”** The treatises now called the *Metaphysics* were Aristotle's “first philosophy”: the study of being as being, and of the highest causes.
- **The four causes** (*Physics* II). A bronze statue has a material cause (the bronze), a formal cause (its shape), an efficient cause (the sculptor) and a final cause (its purpose, such as honouring a god).
- **The eternal world.** Aristotle argued that time and motion have no beginning.
- **The unmoved mover** (*Metaphysics* Λ, Book 12). Everything that moves is moved by something. To avoid an infinite regress of explanation, there must be a mover that is not itself moved: pure actuality, with no unrealized potential. It is “thought thinking itself.” It moves the heavens not by pushing but “as the object of love moves,” as a final cause.

The mover explains *motion* and *order*, not the existence of matter, which is why Aristotle is the textbook Split (SL).

A.5 The Stoics and Plotinus

The Stoics, founded by Zeno of Citium around 300 BCE and including Chrysippus, Seneca, Epictetus and Marcus Aurelius, were materialists. Everything real is a body. They identified God with the active principle in matter: a fiery, intelligent breath (*pneuma*), the logos, which pervades and orders the cosmos. The cosmos periodically dissolves into fire and is reborn in the same order. God and cosmos are two aspects of one physical whole, which is why Stoicism scores E-identity.

Plotinus (204/5–270 CE), the founder of Neoplatonism, taught in Rome. His *Enneads* describe reality as flowing from **the One**, a principle so simple that no predicate applies to it. From the One “overflows” **Intellect** (*nous*, containing the Forms), then **Soul**, then the physical world. Emanation is necessary and involuntary, like light from the sun, yet the One is not diminished. Neoplatonism shaped Augustine, Pseudo-Dionysius, Islamic philosophy and medieval thought.

A.6 Augustine

Augustine of Hippo (354–430 CE) was a North African bishop, a former Manichaean (A.9) and a Neoplatonist before his conversion. In *Confessions* Book XI he asks what God was doing before creating the world, and answers that there was no “before”: God created

time along with the world, and God's eternity is not a very long time but timelessness. He also taught creation from nothing and continuous conservation: if God withdrew his power, creatures would cease to be.

A.7 Avicenna, al-Ghazālī and Maimonides

- **Ibn Sīnā (Avicenna)** (980–1037), a Persian polymath, wrote *The Healing (al-Shifā')*. He distinguished essence (what a thing is) from existence (that it is). A triangle's definition does not tell you whether any triangle exists. Anything whose essence does not include existence is "possible in itself" and needs a cause of its existence. To avoid a regress there must be a **Necessary Existent** (*wājib al-wujūd*), whose essence is its existence. The world flows necessarily from it through a hierarchy of intellects.
- **Abū Ḥāmid al-Ghazālī** (1058–1111), in *The Incoherence of the Philosophers*, attacked Avicenna's necessary emanation and eternal world. In the famous Seventeenth Discussion he argued that fire does not *necessarily* burn cotton. God creates the burning on the occasion of contact, according to God's custom. That is *occasionalism*: God is the direct ground of every regularity.
- **Moses Maimonides** (1138–1204), the Jewish philosopher and legal scholar, wrote *The Guide of the Perplexed*. He argued that neither creation nor the eternity of the world can be demonstrated, but that creation is more plausible and is required by the Torah.

A.8 Thomas Aquinas

Thomas Aquinas (1225–1274), a Dominican friar and theologian, joined Aristotle to Christian doctrine.

- **The Five Ways** (*Summa Theologiae* I, q. 2, a. 3) argue from motion, efficient causation, contingency, degrees of perfection, and the goal-directed behaviour of natural things, to God.
- **God is *ipsum esse subsistens***, subsistent being itself. God's essence is to exist.
- **Creation is not a change** from one state to another. It is the total dependence of a thing's being on God.
- **In *On the Eternity of the World*** (c. 1270) Aquinas argued that it is not contradictory for a created world to have always existed. That the world had a beginning is known by faith, not by reason. The ground/cause-in-time distinction in the infographic is Aquinas's distinction.

A.9 Indian, Chinese and other traditions

- **The Vedas** (c. 1500–500 BCE) are the oldest Sanskrit scriptures. **Ṛta** is the cosmic and moral order. The **Nāsadīya Sūkta** (*Rig Veda* 10.129), the “Hymn of Creation,” begins “Then was neither non-being nor being” and ends in open uncertainty.
- **The Upaniṣads** (c. 800–200 BCE) teach that **Brahman**, ultimate reality, is identical with **ātman**, the deepest self.
- **Sāṃkhya** (classical form in Īśvarakṛṣṇa’s *Sāṃkhyakārikā*, c. 350–450 CE) is dualist and non-theistic. Countless conscious **puruṣas** witness the evolution of **prakṛti**, primal matter made of three strands (*guṇas*).
- **Nyāya-Vaiśeṣika** is the school of logic and atomism. **Udāyana** (c. 975–1050), in the *Nyāyakusumāñjali*, argued that composite things, like pots, need an intelligent maker. So the world needs one: **Īśvara**, who arranges eternal atoms.
- **Mīmāṃsā**, the school of Vedic interpretation (Kumārila Bhaṭṭa, seventh century), argued against a creator. The Veda is eternal and authorless, and the world beginningless.
- **Jainism** (Mahāvīra, sixth to fifth centuries BCE) holds that the universe is uncreated and eternal, and gives explicit arguments against a creator.
- **Buddhism** (the Buddha, fifth century BCE) teaches *pratītyasamutpāda*, dependent origination: everything arises from conditions. The Buddha left “undetermined” (*avyākṛta*) questions such as whether the world is eternal. **Dharmakīrti** (seventh century) argued against Īśvara.
- **Nāgārjuna** (c. 150–250 CE) founded **Madhyamaka**, the “Middle Way.” His *Mūlamadhyamakārikā* argues that everything is empty (*śūnya*) of intrinsic nature (*svabhāva*). Things exist only dependently and conventionally. The *catuṣkoṭi*, or tetralemma, considers four options (is, is not, both, neither) and often rejects all four.
- **Vedānta** comprises the schools interpreting the Upaniṣads, the *Bhagavad Gītā* and the *Brahma Sūtras*. **Śaṅkara** (c. 700–750), Advaita, “non-dual”: only Brahman is ultimately real, and the world is an appearance through *māyā*. **Rāmānuja** (c. 1017–1137), Viśiṣṭādvaita, “qualified non-dualism”: souls and matter are the real body of Brahman, understood as the personal God Viṣṇu. **Madhva** (1238–1317), Dvaita, “dualism”: God, souls and matter are eternally distinct.
- **Sikhism** was founded by Guru Nānak (1469–1539). The *Mūl Mantar* opens the Guru Granth Sahib: *Ik Onkar*, “One Creator,” true, the creator-being, without fear or enmity, timeless, unborn, self-existent.

- **Daoism.** The ***Daodejing*** is attributed to Laozi (compiled c. fourth century BCE). The ***Zhuangzi*** is a book of parables. **Guo Xiang** (d. 312 CE), in his commentary on the *Zhuangzi*, argued that things are “self-so” (*ziran*) and produce themselves, with no creator.
- **Neo-Confucianism.** **Zhu Xi** (1130–1200) synthesized Confucian thought into the orthodoxy of later imperial China. Everything is **li** (principle, pattern) and **qi** (vital stuff, material force). The **taiji**, the Supreme Ultimate, is the sum of all li.
- **Manichaeism** was founded by Mani (216–274/7 CE). Light and Darkness are two eternal principles, and the world is their mixture.
- **Gnosticism** is a family of early Christian-era movements. A lower craftsman, the Demiurge, made the material world, while the true God is a remote Monad.

A.10 Spinoza and Leibniz

Baruch Spinoza (1632–1677), a Dutch-Jewish lens grinder and philosopher, was expelled from his synagogue in 1656. His *Ethics* (published 1677) proceeds “in geometric order,” with definitions, axioms and propositions. There is only one substance, God or Nature, with infinitely many attributes. Everything is a mode of it, and everything follows necessarily from God’s nature. He anticipated a strong form of the PSR.

Gottfried Wilhelm Leibniz (1646–1716), a German polymath, co-invented the calculus. He is known for the **principle of sufficient reason**, for the view that God chose to create the **best of all possible worlds**, and for *monads*, simple mind-like substances. In *On the Ultimate Origination of Things* (1697) he argued that even an infinite series of states of the world, each explained by the previous one, needs an explanation outside the series. His example is an eternal succession of geometry books, each copied from the one before.

A.11 Descartes, Boyle, Newton, the Leibniz–Clarke correspondence and deism

- **René Descartes** (1596–1650) derived the conservation of motion and the laws of motion from God’s immutability (*Principles of Philosophy*, 1644, II.36–37).
- **Robert Boyle** (1627–1691) described laws as God’s rules for matter.
- **Isaac Newton** (1643–1727) argued in the *General Scholium* of the *Principia* (2nd ed., 1713) that “this most elegant system of the sun, planets, and comets could only proceed from the counsel and dominion of an intelligent and powerful being.”
- **The Leibniz–Clarke correspondence** (1715–16) was an exchange between Leibniz and Samuel Clarke, who spoke for Newton. Leibniz mocked a God who must “wind up his watch from time to time.”

- **Deism** was a seventeenth- and eighteenth-century movement, including John Toland, Matthew Tindal and Thomas Paine, that accepted a creator known by reason but rejected revelation and miracles. Its popular image is God as a clockmaker who builds the world and lets it run.

A.12 David Hume

David Hume (1711–1776), the Scottish empiricist, argued that we never observe necessary connections between causes and effects, only constant conjunction. His *Dialogues concerning Natural Religion* (published 1779) stage a debate between Cleanthes (design), Demea (a priori cosmological reasoning) and the skeptic Philo. In Part IX, against Demea, Hume argues that explaining each of twenty particles leaves nothing further to explain about the twenty together. The famous “commit it then to the flames” is the last line of his *Enquiry concerning Human Understanding* (1748).

A.13 Kant and Hegel

Immanuel Kant (1724–1804), in the *Critique of Pure Reason* (1781; 2nd ed. 1787), argued that the mind structures experience through space, time and categories such as causality. These apply only within possible experience. When reason tries to think the world as a completed whole, it produces **antinomies**: equally good proofs that the world has a beginning and that it does not. Kant also argued that the cosmological argument secretly relies on the ontological argument. In the *Critique of Practical Reason* (1788) he held that morality requires us to *postulate* God, freedom and immortality.

G. W. F. Hegel (1770–1831) described reality as the self-development of **Spirit** (*Geist*), which externalizes itself as nature and returns to itself in human history, art, religion and philosophy. “Right” Hegelians read this theistically. “Left” Hegelians, such as Feuerbach and Marx, read it naturalistically.

A.14 Tillich and Heidegger

Paul Tillich (1886–1965), a German-American Protestant theologian, taught in *Systematic Theology* (vol. 1, 1951) that God is not a being alongside others, even the highest, but **being-itself**, the **ground of being**. On his view, to argue that God exists is to make God a thing, and he wrote, provocatively, that “God does not exist,” meaning that God is beyond existence as a being.

Martin Heidegger (1889–1976) opened *Introduction to Metaphysics* (lectures 1935, published 1953) with “Why are there beings at all instead of nothing?” He called it the broadest, deepest and most basic question.

A.15 Whitehead and process philosophy

Alfred North Whitehead (1861–1947), co-author with Russell of *Principia Mathematica*, developed a “philosophy of organism” in *Process and Reality* (1929). Reality consists of momentary “actual occasions” of experience. **Creativity** is the ultimate principle by which many occasions become one new occasion. God has a *primordial nature*, which envisages all possibilities and lures the world toward order and value, and a *consequent nature*, which receives and preserves the world. Charles Hartshorne, John Cobb and David Ray Griffin developed **process theism**.

A.16 Contemporary philosophy of religion

- **Alvin Plantinga** (b. 1932) revived the ontological argument in modal form and defended the rationality of belief in God.
- **Richard Swinburne** (b. 1934) argued in *The Existence of God* that theism is probable on the total evidence, using Bayesian confirmation theory.
- **William Lane Craig** (b. 1949) revived the *kalām* argument. Whatever begins to exist has a cause, the universe began to exist, so the universe has a cause.
- **Alexander Pruss** and **Timothy O'Connor** defend versions of the Leibnizian and necessary-being arguments.
- **Graham Oppy** (b. 1960), an Australian naturalist, compares theism and naturalism as whole worldviews and argues that naturalism is simpler and explains just as much.

A.17 Grounding theory

Since about 2009, metaphysicians have developed a theory of non-causal “in virtue of” dependence.

- **Kit Fine** (“Guide to Ground,” 2012) treats grounding as a sentential operator.
- **Jonathan Schaffer** (“On What Grounds What,” 2009) treats metaphysics as the study of what grounds what, and defends *priority monism*.
- **Gideon Rosen** (2010) connects grounding with reduction and essence.
- **Jessica Wilson** (2014) argues that “Grounding” is too coarse and should give way to specific relations.

A.18 Cosmologists

- **Georges Lemaître** (1894–1966), a Belgian priest and physicist, proposed the “primeval atom,” later called the Big Bang.
- **Edwin Hubble** (1889–1953) measured the expansion.

- **Stephen Hawking** (1942–2018) and **Roger Penrose** (b. 1931) proved the singularity theorems. Hawking and **James Hartle** (1939–2023) proposed the no-boundary wave function.
- **Alan Guth** (b. 1947) proposed inflation.
- **Alexander Vilenkin** (b. 1949) proposed tunnelling from nothing and co-proved the BGV theorem.
- **Max Tegmark** (b. 1967) proposed the mathematical universe hypothesis.

A.19 The Russell–Copleston debate

On 28 January 1948 the BBC broadcast a debate on the existence of God between the philosopher **Bertrand Russell** and the Jesuit historian of philosophy **Frederick Copleston**. Copleston defended Leibniz’s argument from contingency. Russell replied that “the universe is just there, and that’s all,” and that “every man who exists has a mother ... but obviously the human race hasn’t a mother — that’s a different logical sphere.” The debate is reprinted in Russell’s *Why I Am Not a Christian* (1957) and is a standard teaching text.

A.20 Philosophers of language: Frege, Russell, Ramsey, Strawson, Kripke

- **Gottlob Frege** (1848–1925) distinguished a term’s *sense* (its mode of presentation) from its *reference* (what it picks out).
- **Bertrand Russell** (1872–1970), in “On Denoting” (1905), analysed definite descriptions as quantified claims.
- **Frank Ramsey** (1903–1930), a Cambridge prodigy who died at 26, showed how to state a theory’s content without its theoretical terms, in the Ramsey sentence.
- **P. F. Strawson** (1919–2006), in “On Referring” (1950), argued that failed presuppositions leave statements without truth value.
- **Saul Kripke** (1940–2022), in *Naming and Necessity* (1980, from 1970 lectures), argued that names are rigid designators and that there are necessary truths known only by experience.

A.21 Logical positivism, Carnap and Wittgenstein

The **Vienna Circle** of the 1920s and 1930s, including Moritz Schlick and **Rudolf Carnap** (1891–1970), and A. J. Ayer’s *Language, Truth and Logic* (1936) held that a statement is meaningful only if it is true by definition or verifiable by experience. Metaphysics and theology fail both tests. Carnap’s “Empiricism, Semantics, and Ontology” (1950) distinguished *internal* from *external* questions (§8.3).

Ludwig Wittgenstein (1889–1951) ends the *Tractatus* (1921) with the view that what can be said can be said clearly, and “whereof one cannot speak, thereof one must be silent.” Proposition 6.44 separates *how* the world is from *that* it is. Proposition 6.5 says “the riddle does not exist.”

A.22 The revival of metaphysics

- **W. V. O. Quine** (1908–2000), in “On What There Is” (1948), made ontology respectable again as the study of what our best theories quantify over.
- **Kripke** restored talk of necessity and essence.
- **David Lewis** (1941–2001) built a systematic metaphysics that included possible worlds, Humean supervenience, the best-system theory of laws, and the method of defining theoretical terms (§2.2).

A.23 The “why anything?” literature

- **Robert Nozick** (*Philosophical Explanations*, 1981, ch. 2) surveyed possible answers, including a principle of fecundity on which all possibilities are realized.
- **Derek Parfit**’s two-part essay “Why Anything? Why This?” (*London Review of Books*, 1998) separated the two questions and took axiarchism seriously.
- **John Leslie** (*Value and Existence*, 1979) and **Nicholas Rescher** (*Axiogenesis*, 2010) defend axiarchism.
- **Peter van Inwagen** (1996) gave a probabilistic argument that there is something because there are infinitely many ways for there to be something and only one way for there to be nothing.
- **Bede Rundle** (2004) and **Jim Holt**’s *Why Does the World Exist?* (2012) are accessible treatments.

A.24 Idealism, simulation and the mathematical universe

- **George Berkeley** (1685–1753), an Irish bishop, held that to be is to be perceived. Physical objects are collections of ideas, and their regularity is God’s steady will.
- **Bernardo Kastrup** (*The Idea of the World*, 2019) defends analytic idealism: one universal consciousness, of which individual minds are dissociated parts.
- **Philip Goff** defends *cosmopsychism*, the view that the universe as a whole is conscious (*Consciousness and Fundamental Reality*, 2017; *Galileo’s Error*, 2019).

- **Nick Bostrom's** simulation argument (2003) says that at least one of three claims holds: civilizations rarely reach simulation-running capacity, they rarely choose to run simulations, or we are almost certainly simulated.
- **David Chalmers's** *Reality+* (2022) argues that simulated worlds are real.

A.25 Comparative and global philosophy of religion

- **John Hick** (1922–2012), *An Interpretation of Religion* (1989): the great traditions are culturally shaped responses to one transcendent “Real.”
- **Robert Cummings Neville** (b. 1939) directed the Comparative Religious Ideas Project (1995–1999) with Wesley Wildman. It produced three volumes: *The Human Condition*, *Ultimate Realities* and *Religious Truth* (2001).
- **Wesley Wildman** (b. 1961) develops multidisciplinary comparative religious philosophy and “ultimacy models.”
- **J. L. Schellenberg** (b. 1959) proposes **ultimism** and argues that philosophy of religion is at an early stage of inquiry.
- **Yujin Nagasawa** works on perfect-being theism and has led projects in global philosophy of religion.
- **Kevin Schilbrack** argues that philosophy of religion should study all religions and practices, not only theistic propositions.

Appendix B — The original infographic, annotated

The infographic that prompted this thesis, version 1:

God* — a shared definition

Built so rival metaphysical theories can be tested against the same job.

THE DEFINITION

God* is the title of a role, not the name of a being. The role is occupied by whatever, if anything¹, is the non-internal² ground³ of the physical order as a whole⁴.

That whole has two sides. **G-existence:** that there is a physically structured domain at all, rather than no such domain. **G-law:** that this domain is lawful — regular and structured, not a heap of disconnected happenings.

God* names one non-internal ground of both. If only one side has such a ground, or the two grounds are different things, God* is not uniquely filled. Say: existence-ground here, law-ground there.



S SPLIT
two grounds

Existence and lawfulness do not share one outside account.

The domain may be brute and the laws grounded — or the reverse; or two different grounds; or a starter after which law needs no further ground.

MAPS⁵
Some deist pictures · some Platonic pictures (craftsman and Forms as two jobs)

NOTE
If the two grounds differ, do not call both God*. The title is for one ground of both sides.

E EMPTY
no ground

The physical order has no outside account. That includes 'it has always been' and 'it just is.'

Empty accepts the question and answers: nothing fills it. A beginningless or brute physical whole still sits here. Ignorance is not this verdict.

MAPS⁵
Atheism / philosophical naturalism · classical Samkhya · Buddhist schools that deny a creator without treating the question as confused

NOTE
Internal processes that are fundamental among other internal processes do not occupy the role. A later physics theory does not, by itself, fill the role.

R REFUSED
no role

The role itself is the wrong question. There is not a missing occupant; the job description does not apply.

Refused declines the question. Empty says nothing fills the role. Refused says the role misfires. Score the claim, not the tone.

MAPS⁵
Some Madhyamaka readings · some Taoist readings that refuse the Tao as a ground of a whole

NOTE
The definition remains usable as vocabulary even when a tradition rejects the job. Refused is not a fifth kind of occupant.

F FILLED
one ground

One thing outside the physical order is why the order is there and why it is lawful.

One non-internal ground covers both G-existence and G-law. A highest being inside the cosmos does not count. Personhood is an occupant theory, not the role.

MAPS⁵
Judaism · Christianity · Islam · Sikhism · theistic Hinduism (Ishvara; some Vedanta when Brahman grounds the world)

NOTE
Occupant theories are not the role: person, will, goodness, revelation, worship, Trinity, covenant, Quran, avatar. Filled can be personal or not.

INTERNAL VS NON-INTERNAL
Something is internal if stating it already assumes a physical order: spacetime, laws, fields, particles, vacuum, history, or a generator that takes an ordered domain as input. Big Bang, inflation, bounce, quantum gravity, dark energy, and a multiverse generator are internal unless they can be stated without that assumption. An earlier stage of the same order is still internal.

CONSTRAINTS
Ground is not an earlier event in time, not a first particle, and not a piece of the order. Being fundamental among physical processes does not fill the role. Necessity given the laws does not fill the role. Personhood and worship-worthiness are claims about an occupant, not part of the role. Not knowing is unfinished inquiry; it is not Empty and not a fifth verdict.

NOTES ¹ If anything — Empty is a legal answer. ² Internal — already presupposes spacetime, laws, or a physical system.

² Ground — that in virtue of which the whole order is there and lawful, rather than there being no such order; not a cause inside the order. ⁴ Physical order as a whole — G-existence plus G-law.
³ Tags show where a tradition often lands, not a claim about every adherent.

The table goes through it element by element. It says what works, what the thesis changed, and why.

Element	What works	What v2 changes, and why
Title and subtitle, “Built so rival metaphysical theories can be tested against the same job”	States the purpose exactly: an instrument, not a doctrine	No change
Definition sentence	Title/role framing (Pike, Lewis); “if anything” allows Empty	“ground” becomes ground* (Ch. 9); “physical order” becomes P* (Ch. 12); the terminal clause is added (§9.3)
G-existence and G-law	Two explananda with deep historical roots (Ch. 3)	Both include specific content: which inventory, which laws (§13.3). A note that their independence depends on the theory of laws (§4.3).
Gold italic note on Split	Anticipates the Split verdict and names it well	No change
Keystone arch diagram	Apt: a structural, non-temporal ground that locks both sides	Optional: draw Empty as a corbelled arch (§6.4)
Split panel	The only framework known to the author with a named verdict for this	Map: move classical deism to Filled; add Aristotle, Nyāya, process theism (§10.9–10.11). Add the subtypes.
Empty panel	“Ignorance is not this verdict”; the note that fundamental is not a ground	Add the subtypes brute, necessary, regress, identity (§10.4). Add the Stoics as an example of Empty with a deity.

Element	What works	What v2 changes, and why
Refused panel	“Score the claim, not the tone”; Refused is not an occupant	Add the three presuppositions and the $R \rightarrow E$ notation (Ch. 11). Add early Buddhism’s undetermined questions and Carnap.
Filled panel	Occupant theories kept out of the role; “Filled can be personal or not”	Add impersonal and non-theistic examples: Plotinus, Spinoza, axiarchism, Tillich
Internal vs non-internal box	The right idea, with good examples	Use the (a)–(c) test; say that a simulator is inside P^* ; rename “internal” to “intra-physical” in formal use (§8.3)
Constraints box	The ground-versus-earlier-event rule (Aquinas) and “not knowing is not a verdict”	Add: once or continually is a mode, not a verdict; a practice that makes no claim is not scored
Footnotes 1–5	Footnote 5, that tags are not claims about every adherent, is methodologically important	Add footnotes for ground*, P^* , and “Empty is not śūnyatā”

Overall. The infographic is a strong first draft of a real idea. Its central moves are sound: title not name, thin role, two sides, four verdicts, Refused distinct from Empty, and occupant theories kept out. Every problem found in Part III is a matter of *precision*, not of direction.

Appendix C — Glossary

Term	Meaning
Apophatic theology	Speaking of God by saying what God is not
Archē	Greek: origin or first principle
Asymmetric	If A relates to B, B does not relate to A in the same way

Term	Meaning
Avyākṛta	Sanskrit: questions the Buddha left undetermined, such as whether the world is eternal
Axiarchism	The view that the world exists because it is good that it should
Brute fact	A fact with no explanation
Catuṣkoṭi	The tetralemma: is, is not, both, neither
Contingent	True, but could have been false
Cosmological argument	An argument from the existence of the world to a first cause or necessary being
Creatio ex nihilo	Creation from nothing: no pre-existing material
Deism	Belief in a creator known by reason, without revelation or intervention
Demiurge	Plato's divine craftsman in the <i>Timaeus</i>
Emanation	The flowing of reality from a source, as in Plotinus
Equivocation	Using one word in two senses within one argument
Explanandum and explanans	What is explained, and what explains it
Fine-tuning	The apparent sensitivity of life to the values of physical constants
G-existence	The fact that there is a physical order at all, with its inventory and initial state
G-law	The fact that the physical order is lawful, with the laws it has
God*	The title of the role "terminal non-internal ground* of P"
Ground*	Generic asymmetric explanatory dependence: productive, constitutive, nomic or axiological
Grounding	In metaphysics, non-causal "in virtue of" dependence

Term	Meaning
Hempel's dilemma	"Physical" defined by current physics is false; by ideal physics, vague
Humean supervenience	Everything supervenes on the local mosaic of particular facts
Hyperintensional	Distinguishing between necessarily equivalent items
Immanent	Within, not beyond
Internal (intra-physical)	Part of P*, stated in P*'s vocabulary, or identical to P* or an aspect of it
Kalām	An Islamic tradition of theology, and the argument from the universe's beginning
Li and qi	Neo-Confucian principle or pattern, and material force
Logos	Greek: word, reason, rational order
Māyā	In Advaita, the power by which Brahman appears as the world
Modal collapse	The PSR seeming to make every truth necessary
Necessary being	Something that exists in every possible world
Occasionalism	The view that God is the only true cause, and creatures are occasions
Occupant theory	Claims about what fills the role: personal, good, and so on
P*	The maximal physical order: everything linked to paradigm physical objects by spatiotemporal relations or law-governed interaction
Pantheism and panentheism	God is the world, or the world is in God but God exceeds it
Plural logic	Logic that refers to many things at once without forming a set
Possible world	A complete way reality could have been

Term	Meaning
Presupposition	What a question or statement takes for granted
Principle of sufficient reason (PSR)	Everything has an explanation
Ramsey sentence	A theory restated with its theoretical terms replaced by bound variables
Rigid designator	A term that picks out the same thing in every possible world
Ṛta	Vedic cosmic order
Svabhāva	Sanskrit: intrinsic, independent nature
Terminal clause	The occupant is the ground that no further non-internal item grounds
Title term	A term naming an office, not an individual
Unmoved mover	Aristotle's eternal final cause of motion
Verdict profile	A person's degrees of confidence across F, S, E and R

Appendix D — From this document to a paper or a doctorate

D.1 An honest assessment

This document is *not* a doctoral thesis. A thesis is examined original research, written over several years under supervision, and the degree is awarded by a university. What it does is show the shape such a project could take. The core idea is **publishable as a journal article** if it is written tightly (8,000–10,000 words), engages the literature in Chapter 8, and argues for the rulings in Part III rather than merely stating them.

The most defensible claim of originality is the *instrument*: a two-sided role, four verdicts with a presupposition test, and the scoring protocol. It is not a claim to have discovered a new theory of God.

D.2 Path 1: a journal article

A suggested structure, around 9,000 words:

1. Introduction: the equivocation problem, with the Stoic and Tillich examples (800 words).
2. The role: title terms, Ramsey–Lewis, the thin role (1,200).
3. Two sides and ground*: why the strict grounding reading and the causal reading are both biased (1,800).
4. Four verdicts and the presupposition test (1,500).
5. Hard cases: simulation, Spinoza, a necessary cosmos, Advaita, axiarchism (2,000).
6. Comparison with Pike, Neville and Wildman, Hick, Schellenberg, Oppy (1,000).
7. Objections and conclusion (700).

Journals to consider. *Religious Studies*, *Sophia*, *International Journal for Philosophy of Religion*, *European Journal for Philosophy of Religion*, *Faith and Philosophy*, *Philosophy East and West* for the comparative angle, and *Journal of the American Academy of Religion* for the religious-studies angle. Check each journal's current scope and submission guidelines before sending.

How submission works.

- Most philosophy journals use anonymous review. Remove your name, and submit to one journal at a time.
- Decisions often take three to nine months.
- Rejection, or “revise and resubmit,” is the normal outcome, even for experienced authors. Referee reports are valuable free feedback.
- Post a preprint on PhilArchive (philarchive.org) to establish priority and invite comment.
- Independent scholars can and do publish. Clarity and engagement with the literature matter more than affiliation.

D.3 Path 2: doctoral study

- **Requirements.** Usually a bachelor's degree, often a master's. Admissions typically ask for a writing sample, which is where a polished version of the God* paper would go. They also ask for a statement of purpose and letters of recommendation.
- **Where.** Look for departments strong in philosophy of religion, metaphysics or comparative philosophy. Boston University (the home of the Comparative Religious Ideas Project), the University of Hawai'i at Mānoa (known for comparative philosophy)

and the University of Birmingham in the UK (global philosophy of religion) are examples. Many others have relevant faculty. Read recent faculty publications before applying.

- **A bridge.** A master’s degree in philosophy or religious studies, or taking individual courses, builds the background and the letters that doctoral admission needs.
- **Funding.** In the US, many philosophy PhD programmes are funded through teaching assistantships. Elsewhere, funding varies widely.

D.4 Path 3: public and collaborative routes

- A public essay, of 2,500–4,000 words, for a general-audience outlet on ideas.
- A talk or poster at the American Academy of Religion, the British Society for the Philosophy of Religion, the European Society for Philosophy of Religion, or an APA divisional meeting.
- Emailing a two-page summary plus the infographic to a researcher whose work you cite, asking a specific question. Specific questions get answers. “What do you think?” often does not.
- Publishing the atlas as an open, correctable resource, with insiders from each tradition invited to challenge the maps.

D.5 A twelve-month plan

Months	Goal	Output
1-2	Read the tier-1 and tier-2 texts below	Notes on each neighbour in Chapter 8
3-4	Draft the article	First full draft
5	Get feedback from two or three readers, including one outside philosophy of religion	Revised draft
6	Preprint on PhilArchive; submit to a first-choice journal	Submission
6-12	Present the idea publicly; revise the infographic to v2; build out the atlas	Public essay, talk, v2 infographic
9-12	Respond to the referee reports	Resubmission, or submission to the next journal

D.6 A reading ladder

- **Tier 1, start here.**
 - Holt, *Why Does the World Exist?*
 - The Russell–Copleston debate.
 - Plato, *Timaeus* 27d–34b.
 - The Stanford Encyclopedia of Philosophy entries on the cosmological argument, grounding, laws of nature, and physicalism.
 - **Tier 2.**
 - Pike (1969).
 - Lewis (1970).
 - Parfit (1998).
 - Oppy, *The Best Argument against God*.
 - Swinburne, *The Existence of God*, chapters 7–8.
 - Schellenberg, *Prolegomena*, chapter 1.
 - Hick, *An Interpretation of Religion*, part 4.
 - **Tier 3.**
 - Fine, “Guide to Ground.”
 - Wilson, “No Work for a Theory of Grounding.”
 - Pruss, *The Principle of Sufficient Reason*.
 - Neville (ed.), *Ultimate Realities*.
 - Wildman, *In Our Own Image*.
 - Garfield’s translation of Nāgārjuna.
 - Deutsch, *Advaita Vedānta*.
 - Chan, *A Source Book in Chinese Philosophy*, the Zhu Xi chapters.
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